

The 8th WMO Symposium on Data Assimilation &
12th International Symposium on Data Assimilation

An Ensemble Kalman Smoother for Online Paleoclimate Data Assimilation

Haohao Sun¹, Lili Lei¹, Zhe-Min Tan¹, Liang Ning², and Zhengyu Liu³

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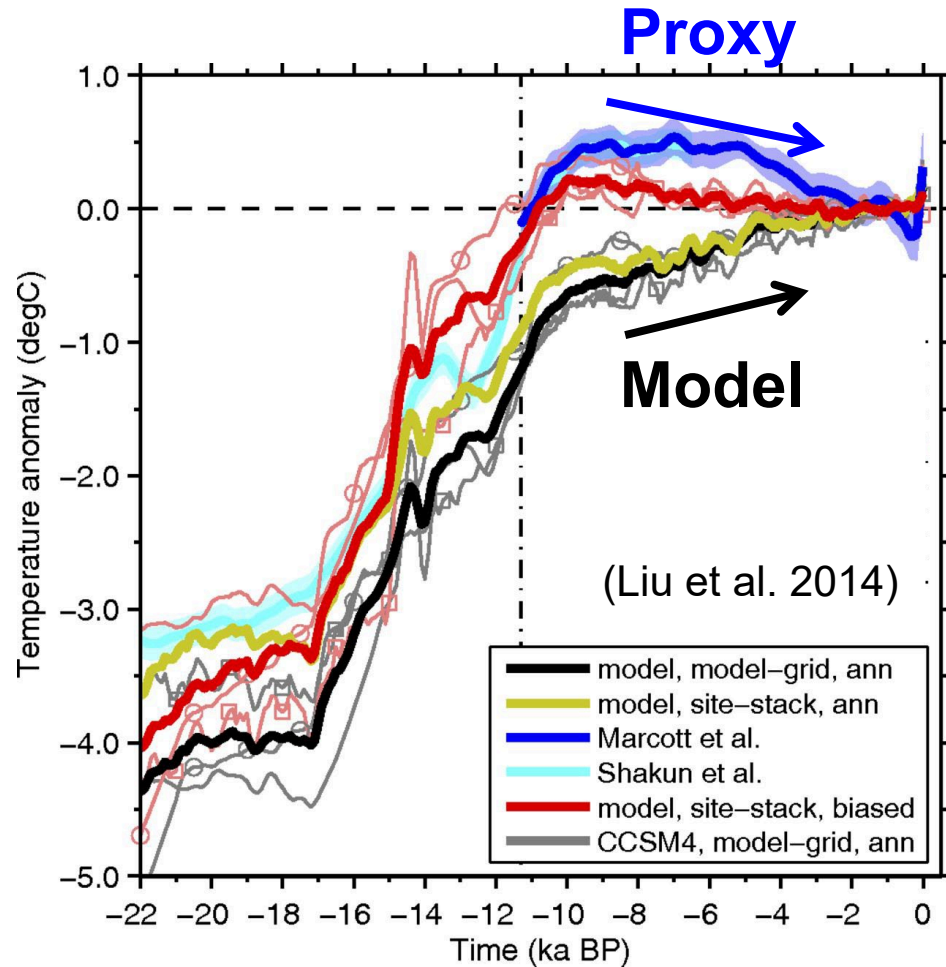
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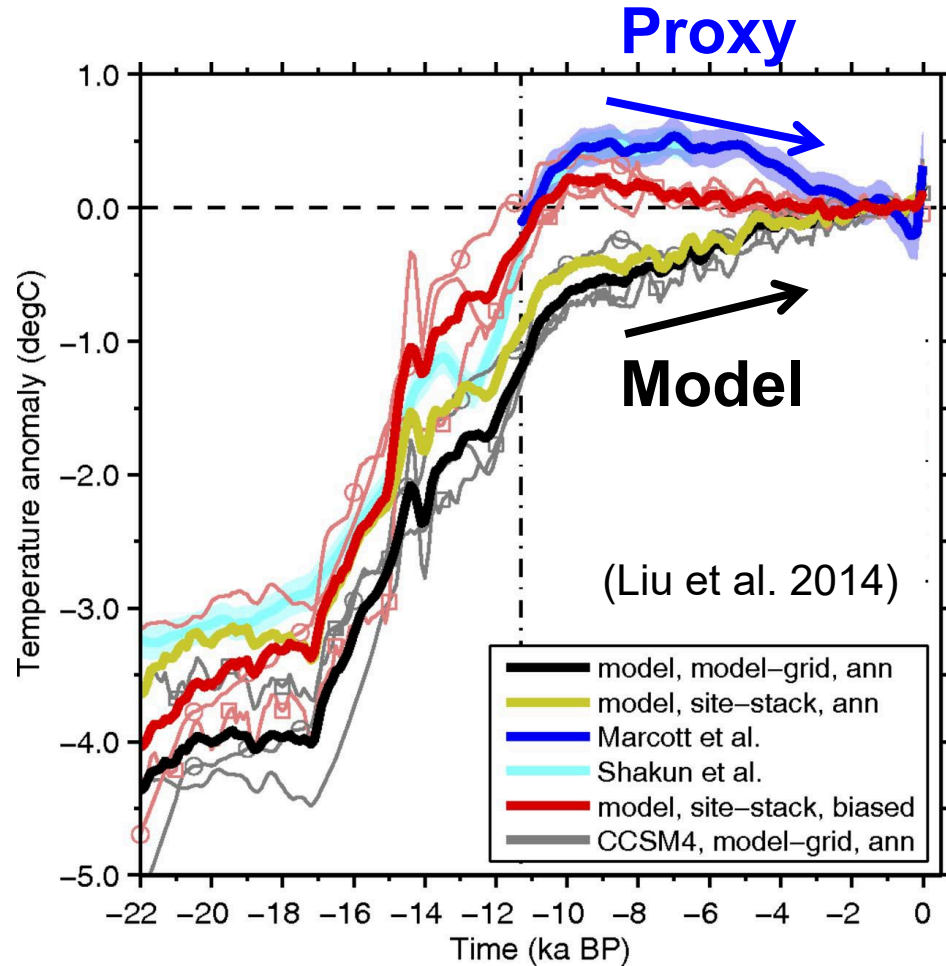
Why paleoclimate needs data assimilation?



Holocene temperature conundrum:

From ~10000 years ago to present,
temperature $\uparrow$ based on the model
temperature $\downarrow$ based on the proxy

Why paleoclimate needs data assimilation?



Holocene temperature conundrum:

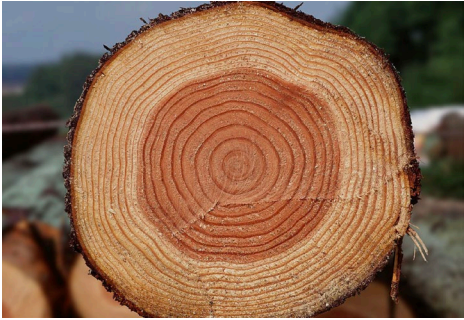
From ~10000 years ago to present,
temperature $\uparrow$ based on the model
temperature $\downarrow$ based on the proxy

Paleoclimate data assimilation (PDA):

combines information from both proxy and climate model

Multi-source proxies

Tree rings



Corals



Ice cores



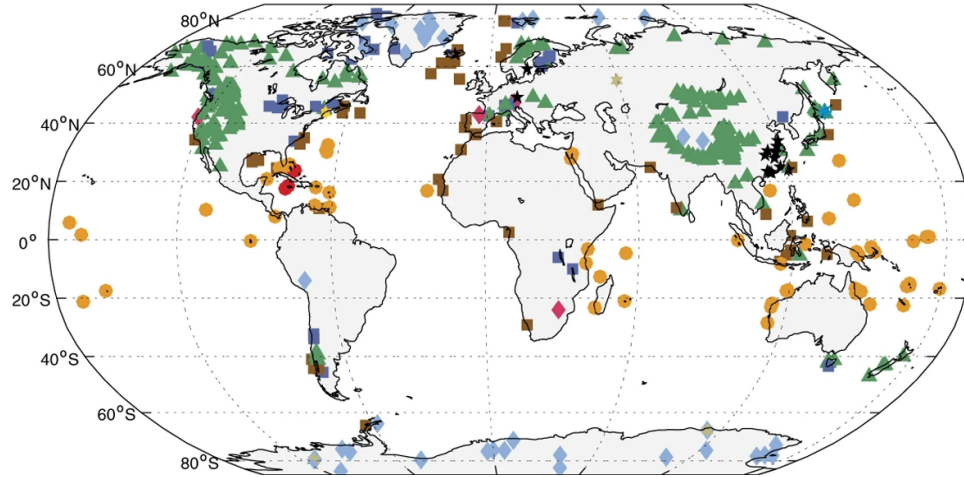
Sediments



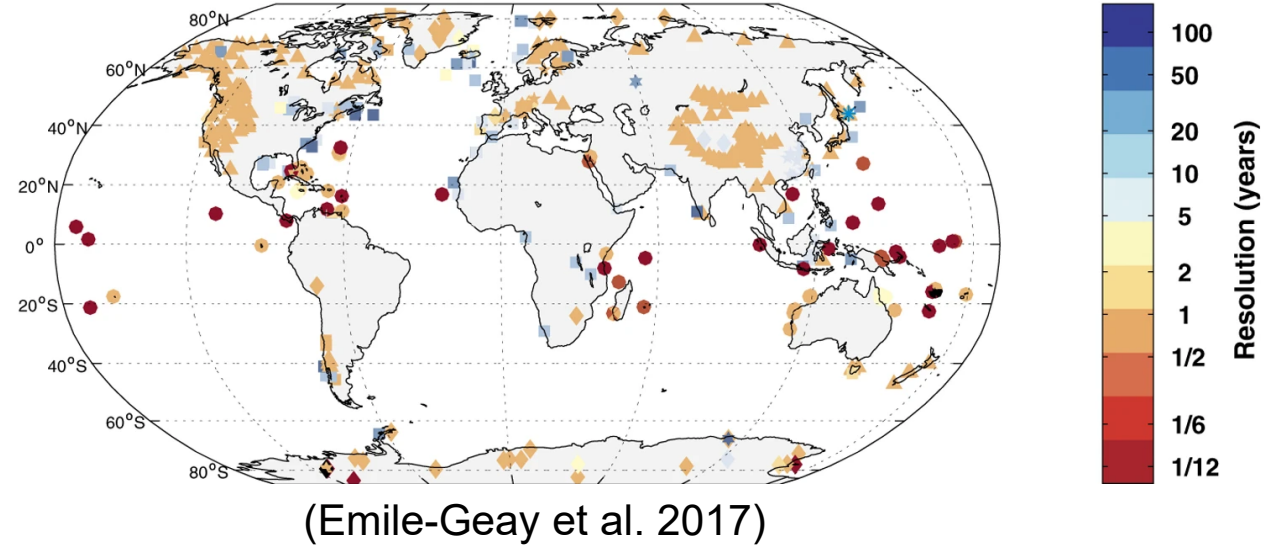
- Growth of trees is controlled by temperature and moisture
- Growth of corals is sensitive to sea surface temperature (SST)
- Atmospheric components of the past are kept in air bubbles in ice cores
- Isotopes in ocean sediments reflect the strength of ocean currents

Multi-scale proxies

Spatial distribution



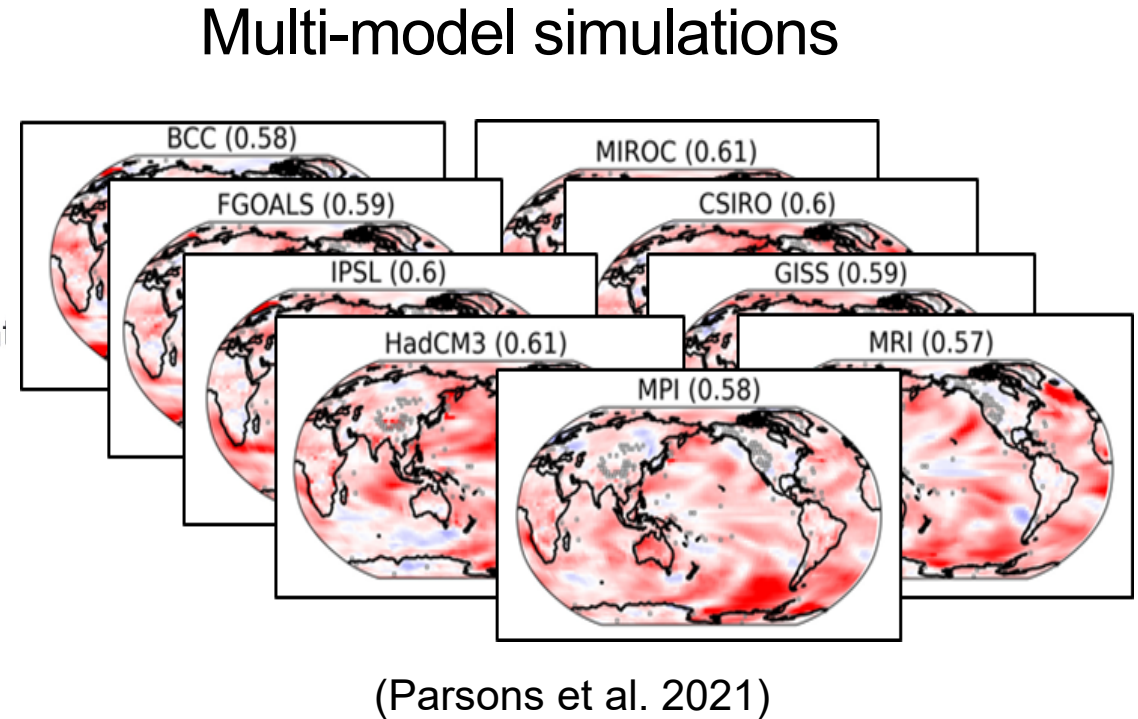
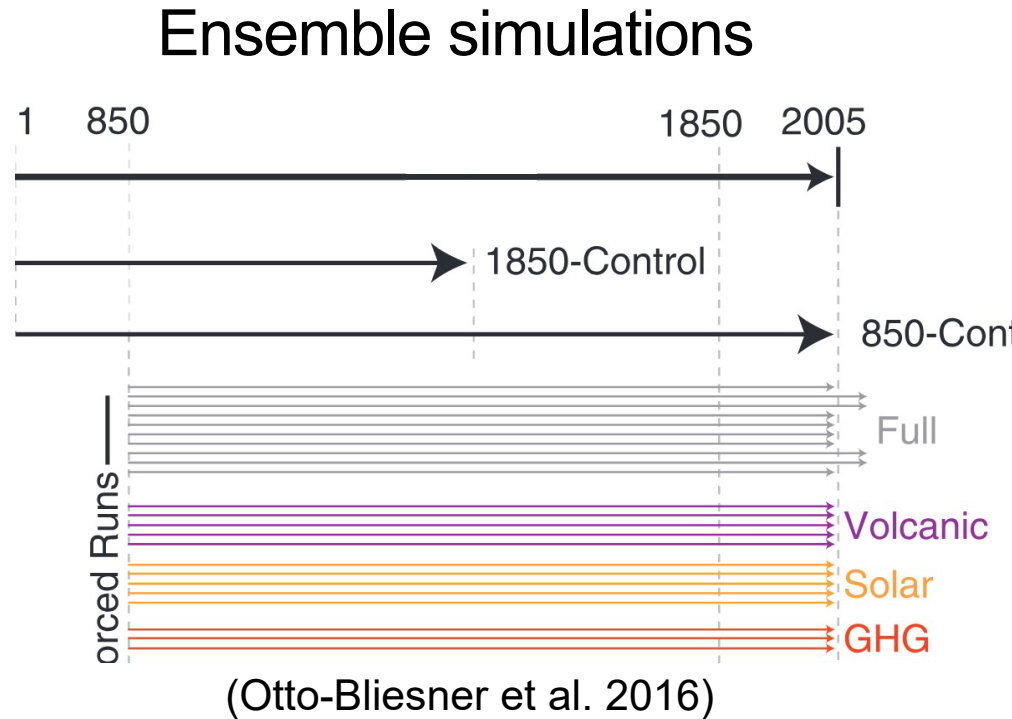
Temporal resolution



(Emile-Geay et al. 2017)

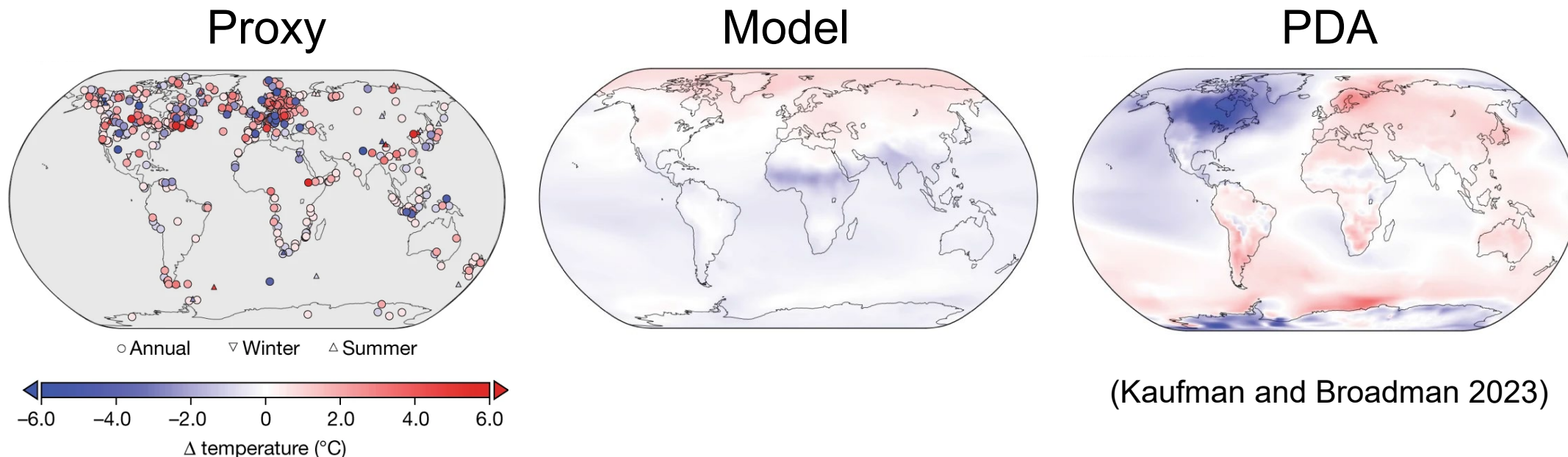
- Proxies are sparse and unevenly distributed globally
- Temporal resolutions span from months to centuries

Multi-ensemble simulations



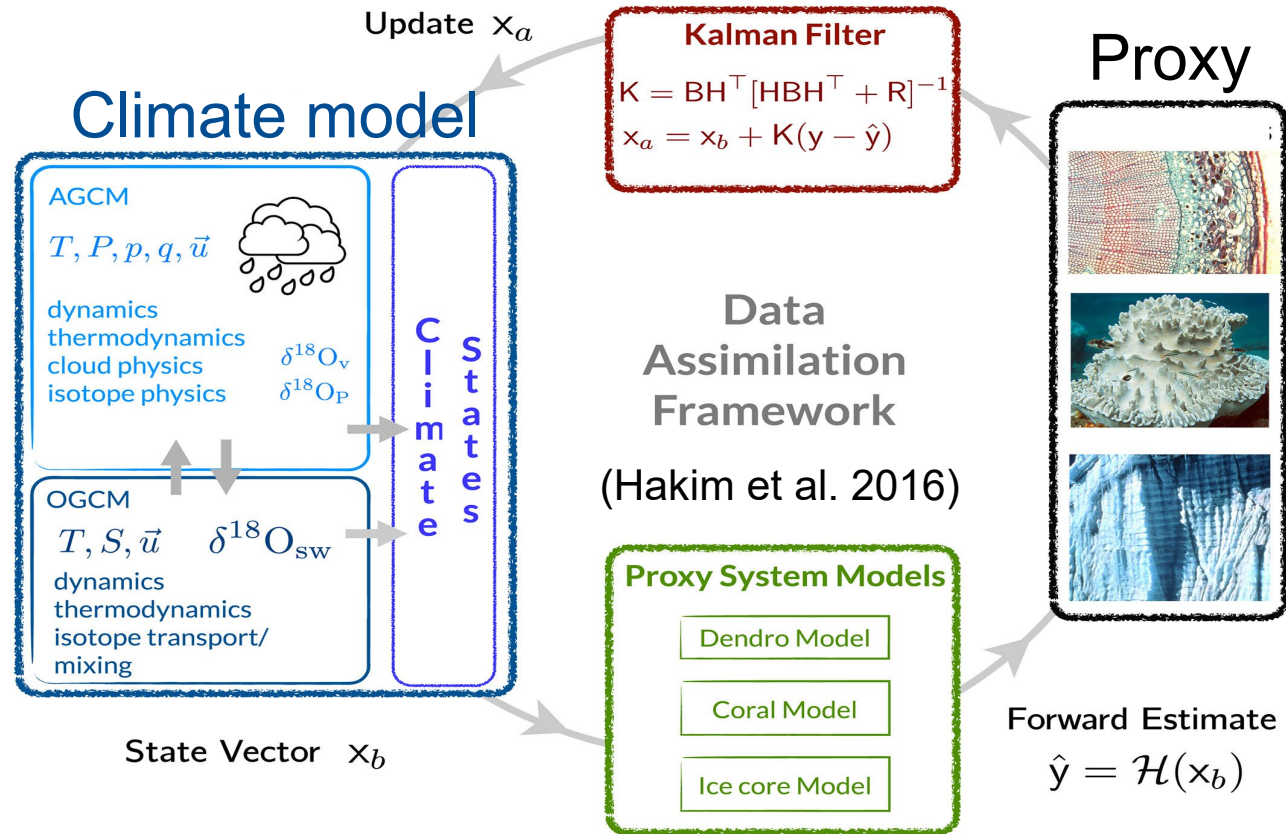
- **Single-model ensembles:** different initial conditions and forcings
- **Multi-model ensembles:** different dynamical cores, parameterizations, and model structures

Paleoclimate Data Assimilation (PDA)

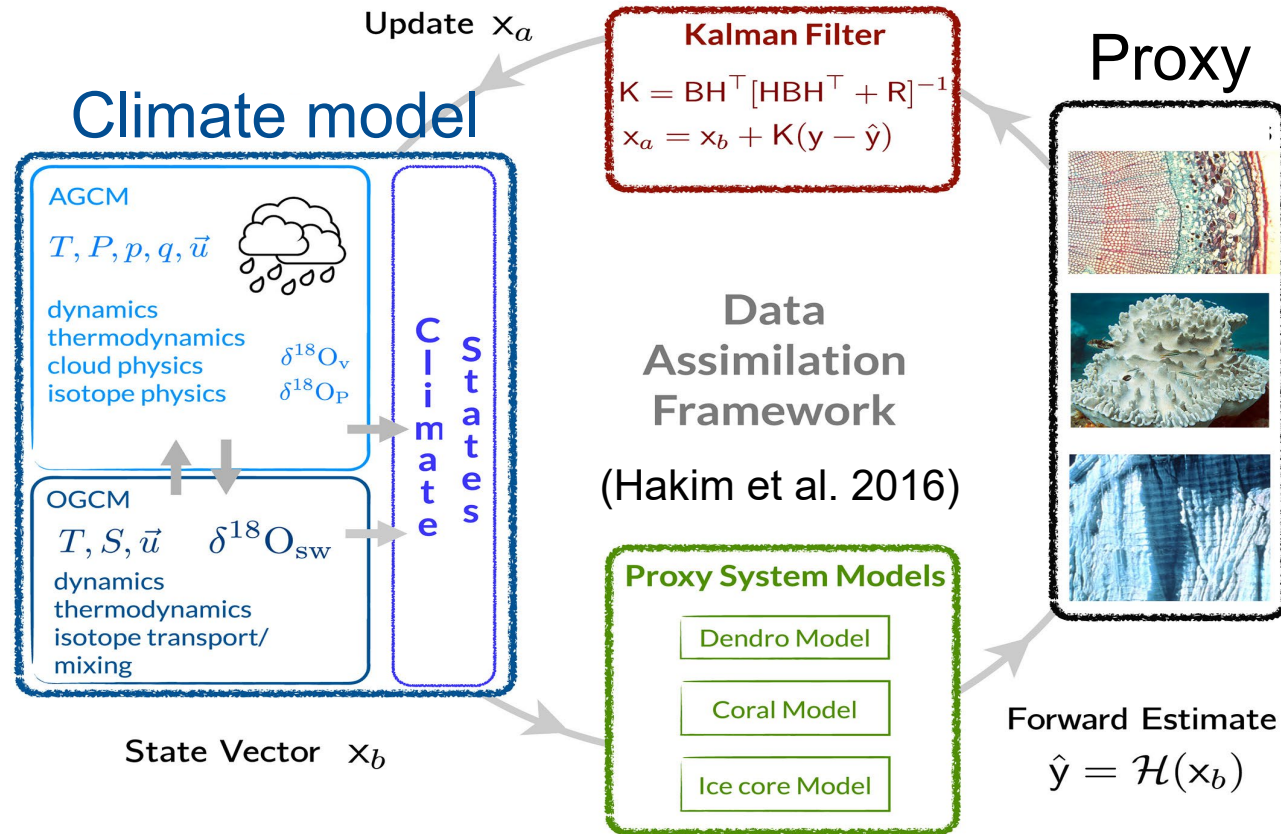


- **Proxy:** **real**-world records, but sparse and low-resolution
- **Model:** **high-resolution**, but imperfect representations of reality
- **PDA:** **realistic, high-resolution** optimal reconstruction of the past

Ideal PDA framework

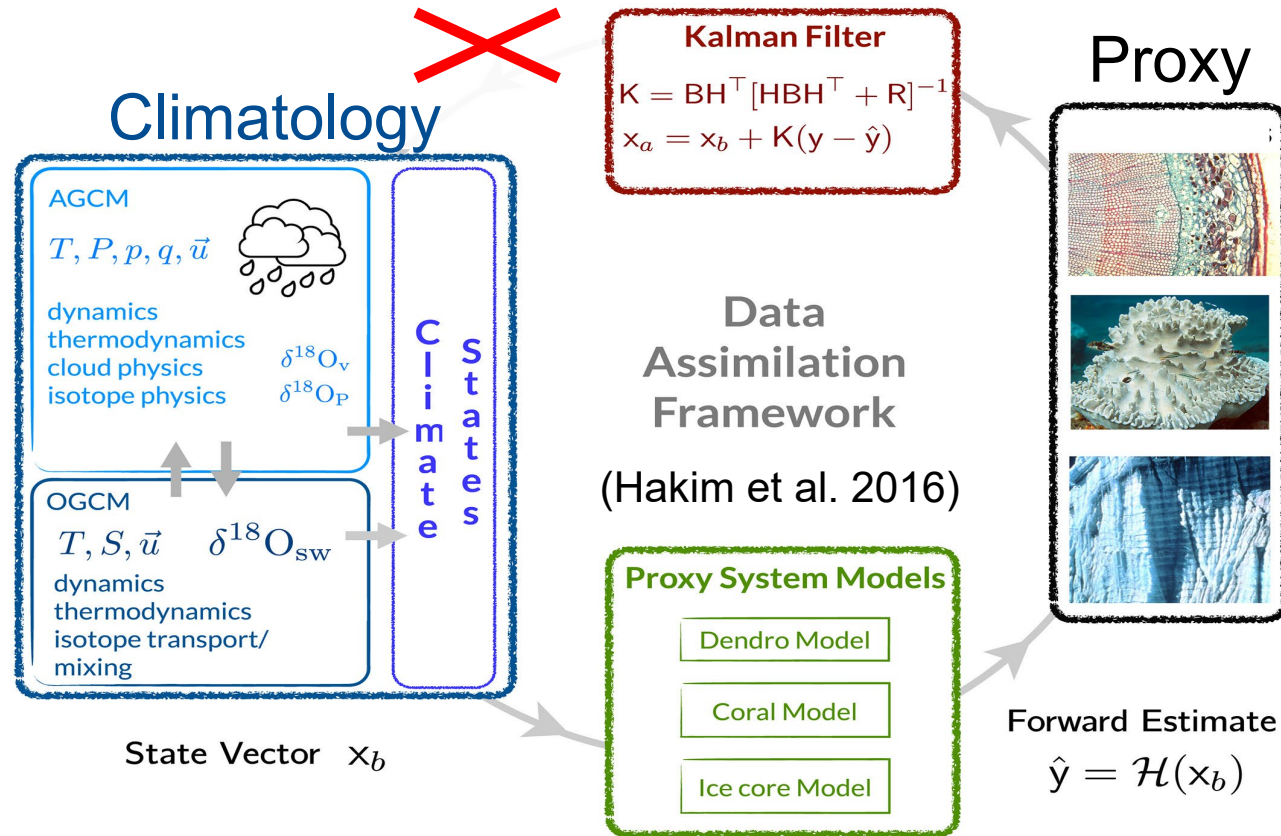


Ideal PDA framework



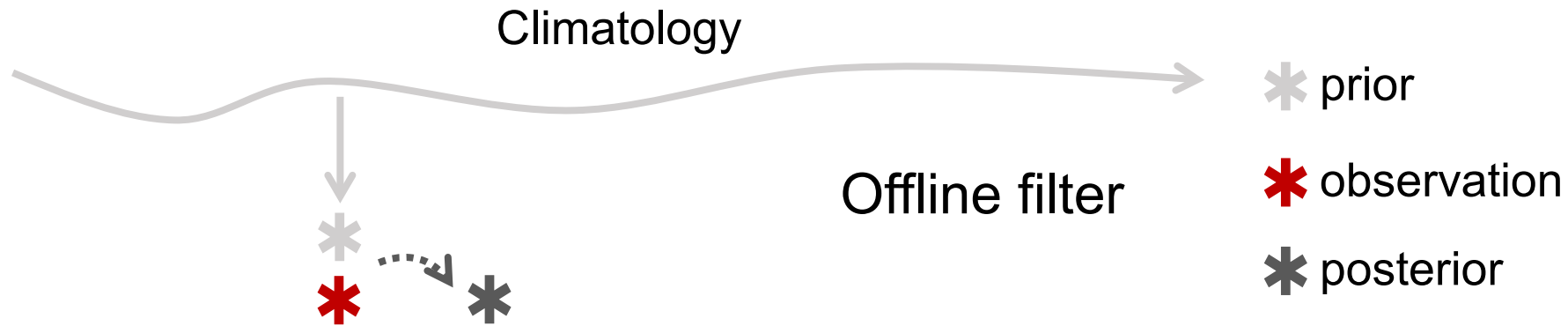
- Expensive long-term model integrations
- Limited constraints from sparse proxies

Offline PDA



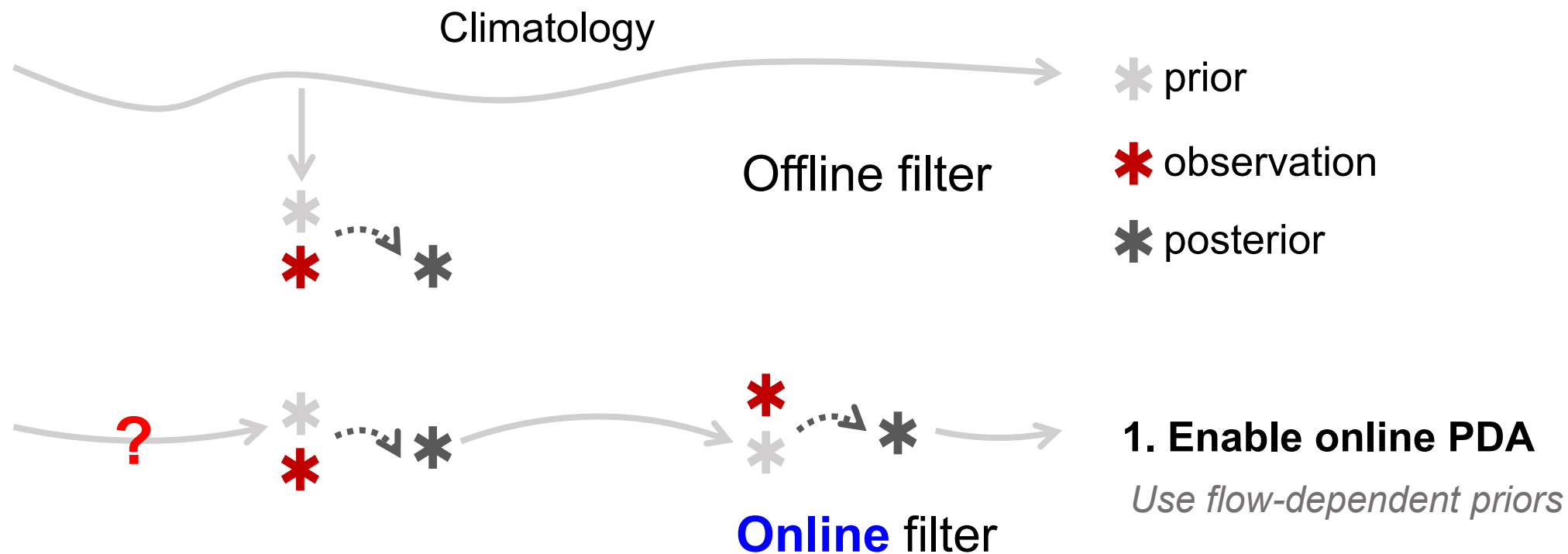
- Expensive long-term model integrations
- Limited constraints from sparse proxies
- Existing simulations $\rightarrow$ climatological priors $\rightarrow$ offline PDA

Challenge in PDA

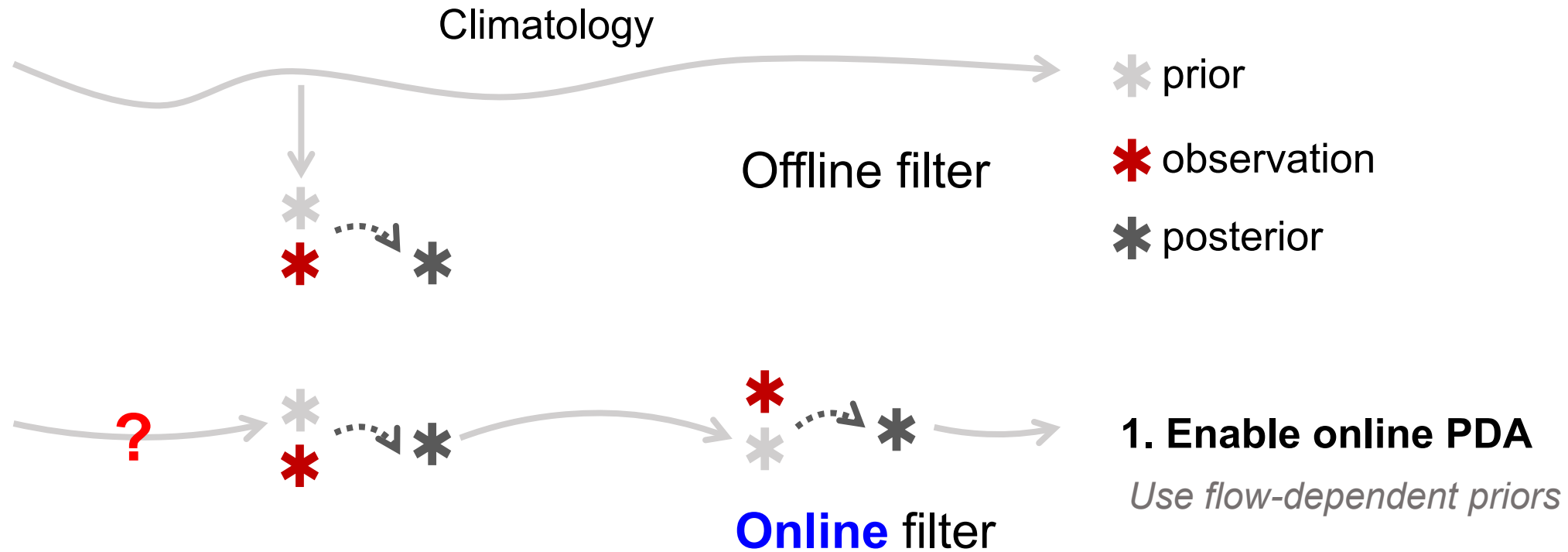


- Evolving climate requires flow-dependent priors

Challenge in PDA

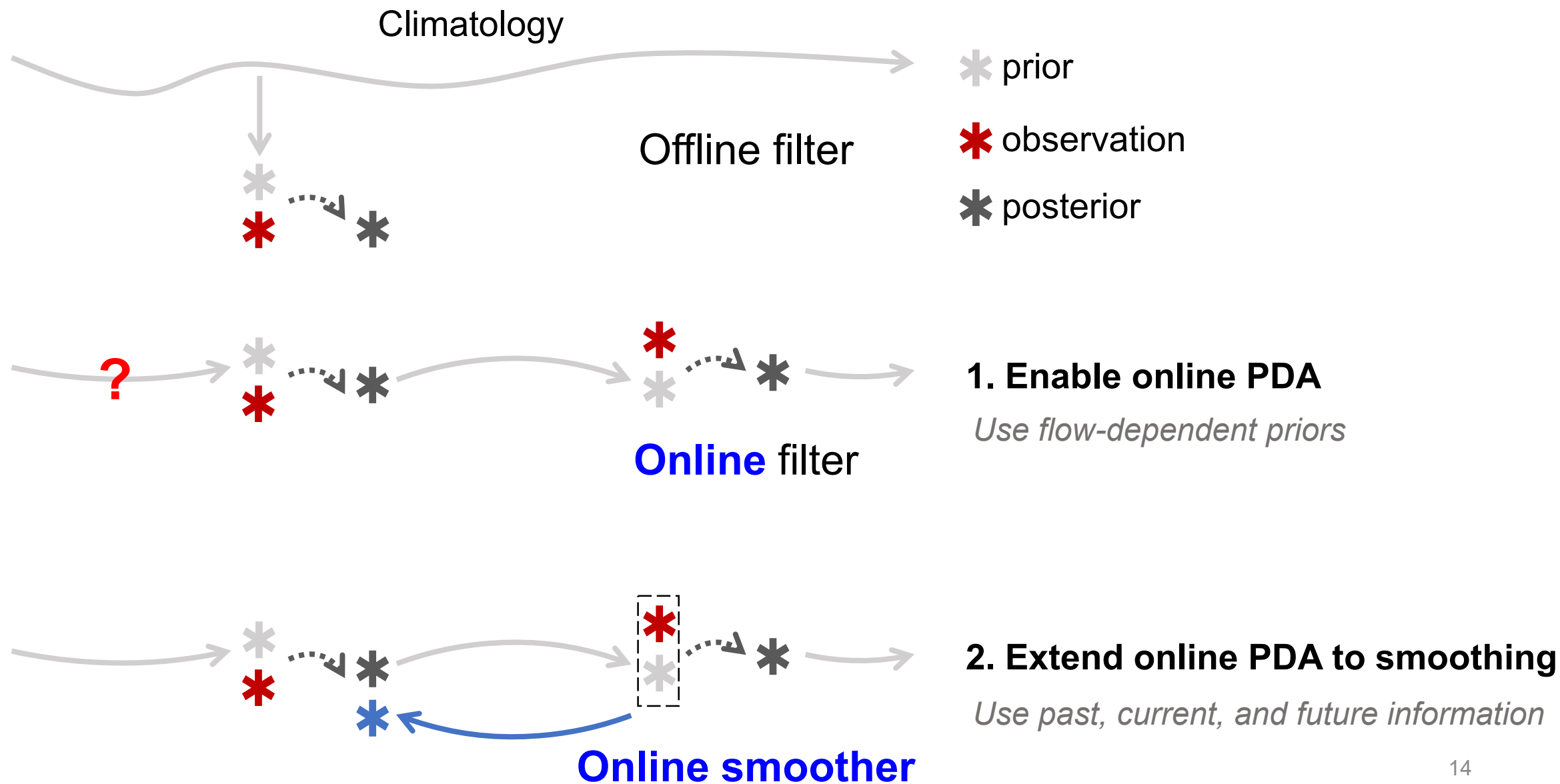


Challenge in PDA



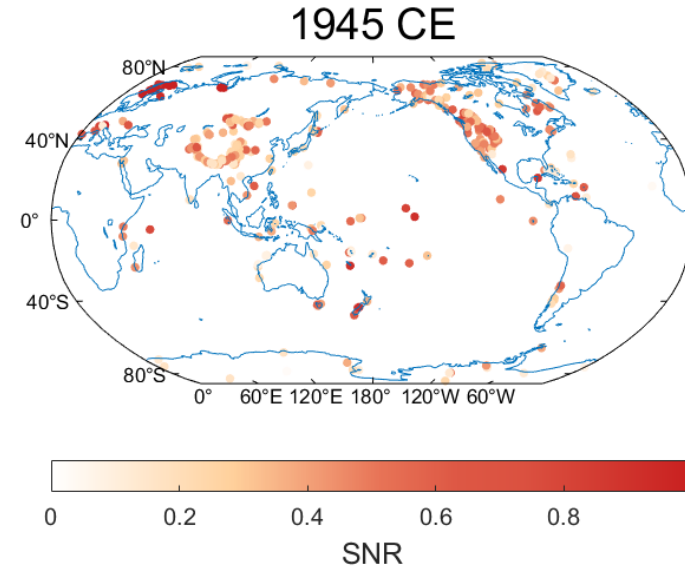
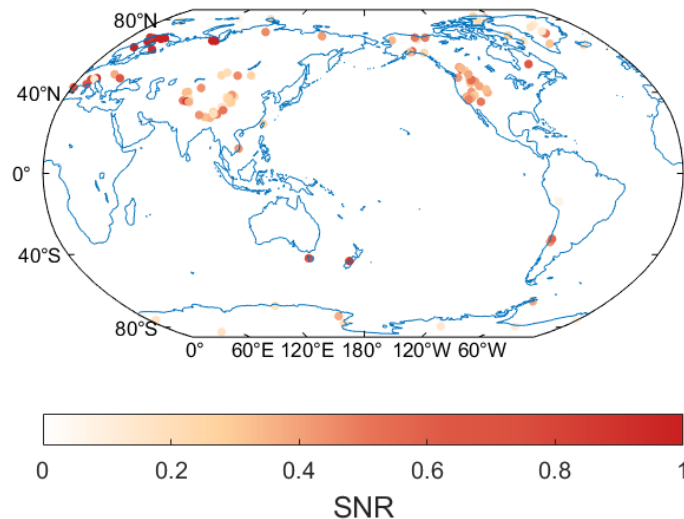
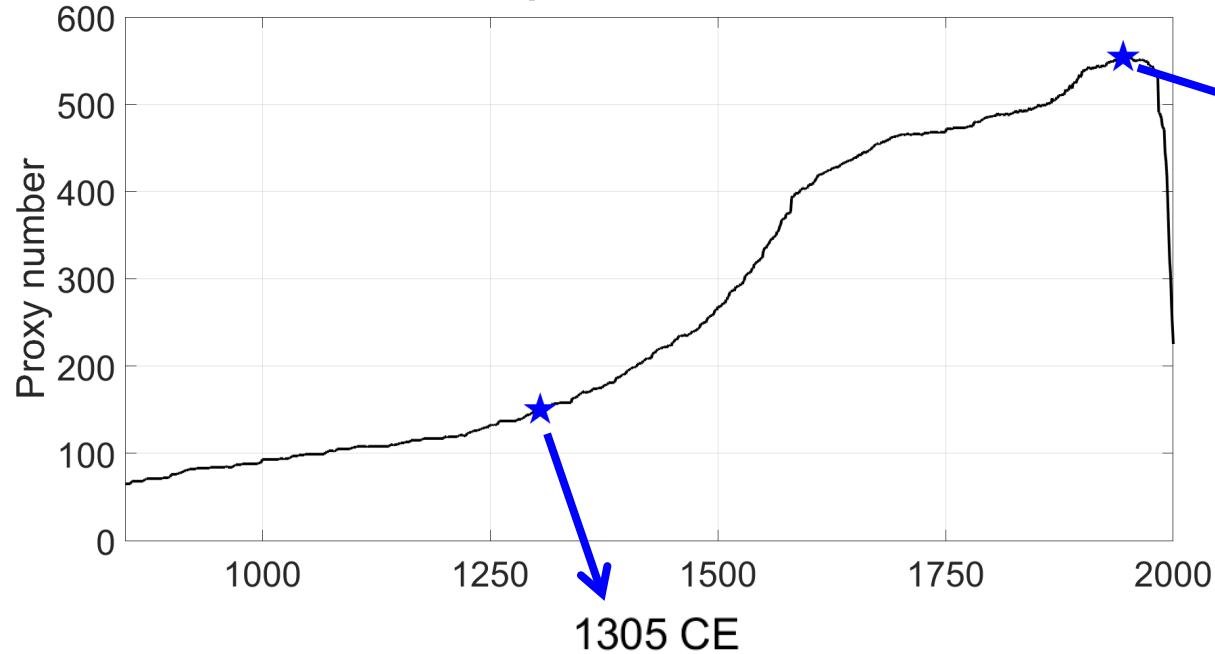
- **Weather DA:** forecast problem — future observations unavailable;
PDA: reconstruction problem — future proxies available

Challenge in PDA



Last millennium proxy

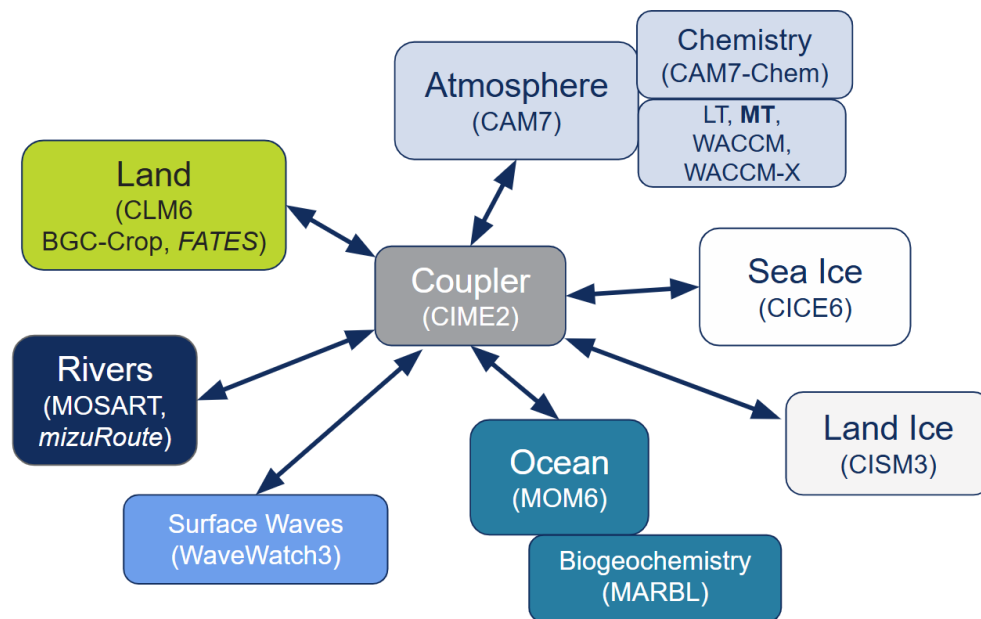
Temporal distribution



- The number of proxies decreases rapidly back in time
- Annual and seasonal proxies with reliable dating are selected for assimilation

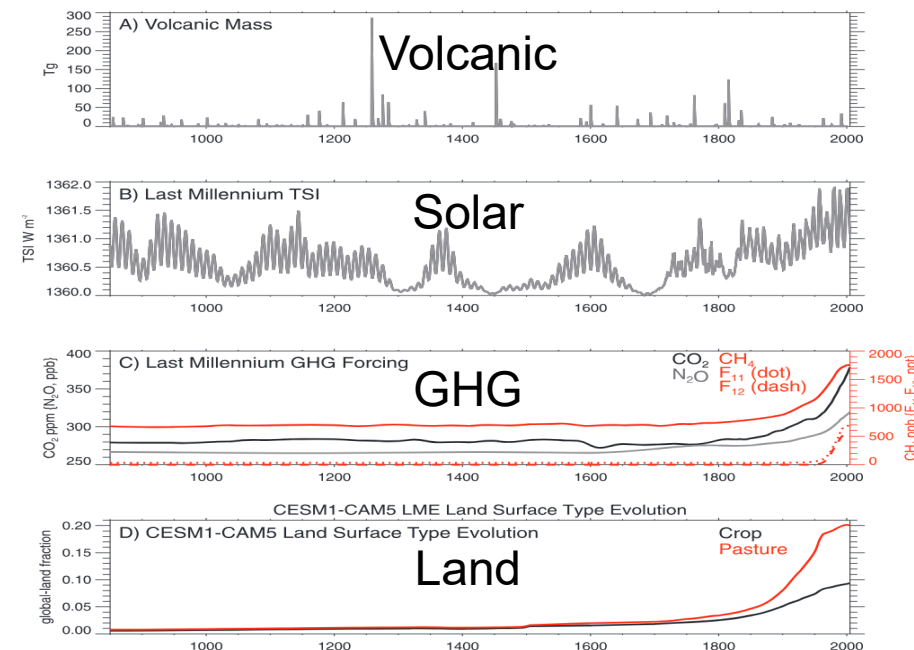
Last millennium model simulation

Community Earth System Model (CESM)



(David et al. 2025)

Forcings

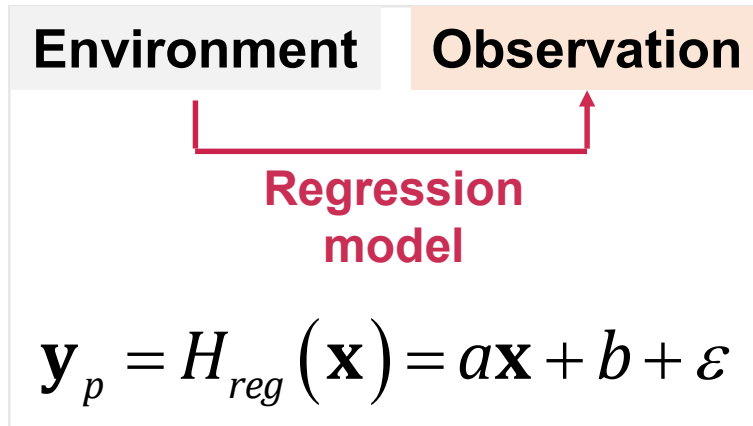


(Otto-Bliesner et al. 2016)

- CESM couples components from atmosphere, ocean, land, sea ice
- Last Millennium Ensemble (LME) is driven by solar, volcanic, orbital, land use/cover, greenhouse gases

Proxy System Model (PSM): from climate state to proxy

Statistical PSM



(Tardif et al. 2019)

Proxy's seasonality

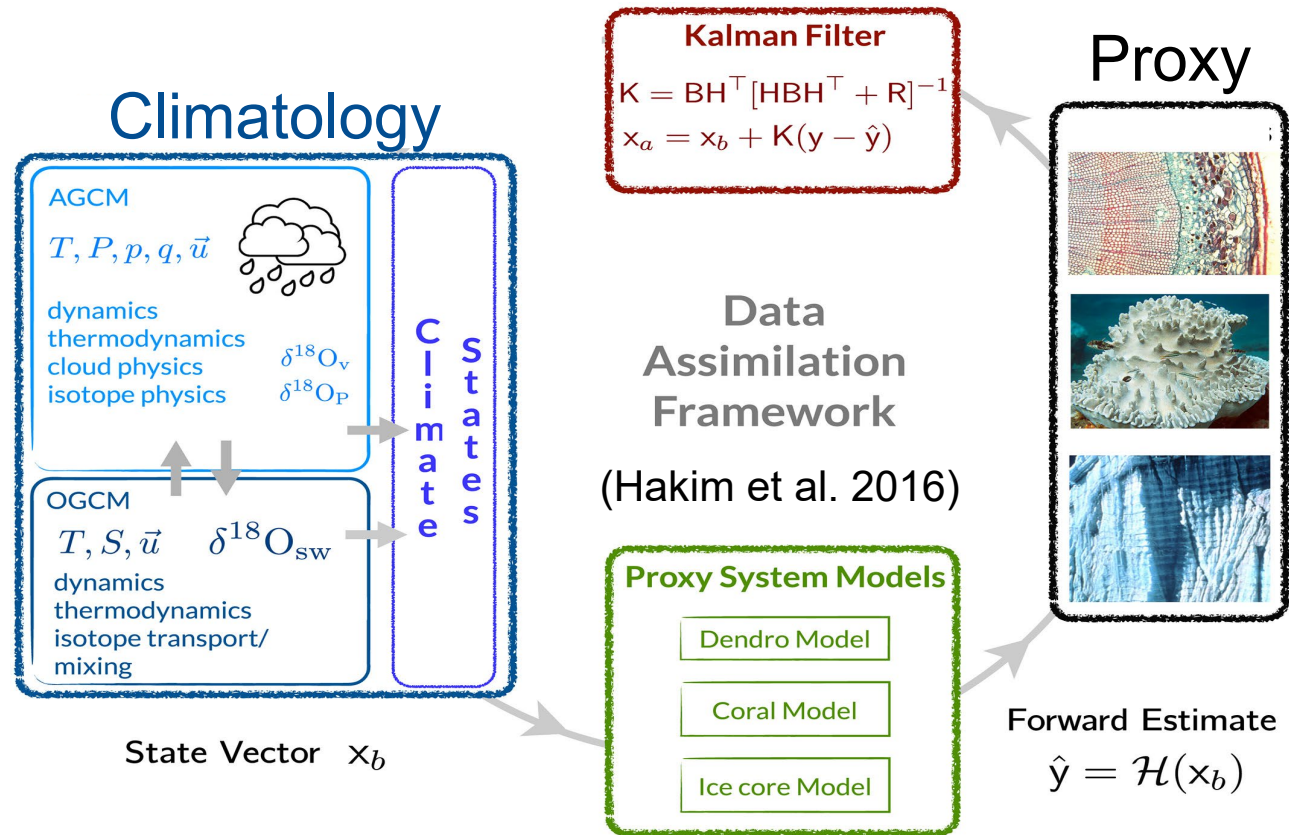
Seasonality	Annual	JJA	MAMJJA	JJASONDJF	SONDJF	DJFMAM	
Number of SAT proxies	78	169	74	49	13	55	22
Number of SST proxies	94	/	/	/	/	/	/

- Statistical PSM: simple, flexible, with explicit error estimate($R = var(\varepsilon)$)
- Proxy's seasonality: determined by maximum explained variance

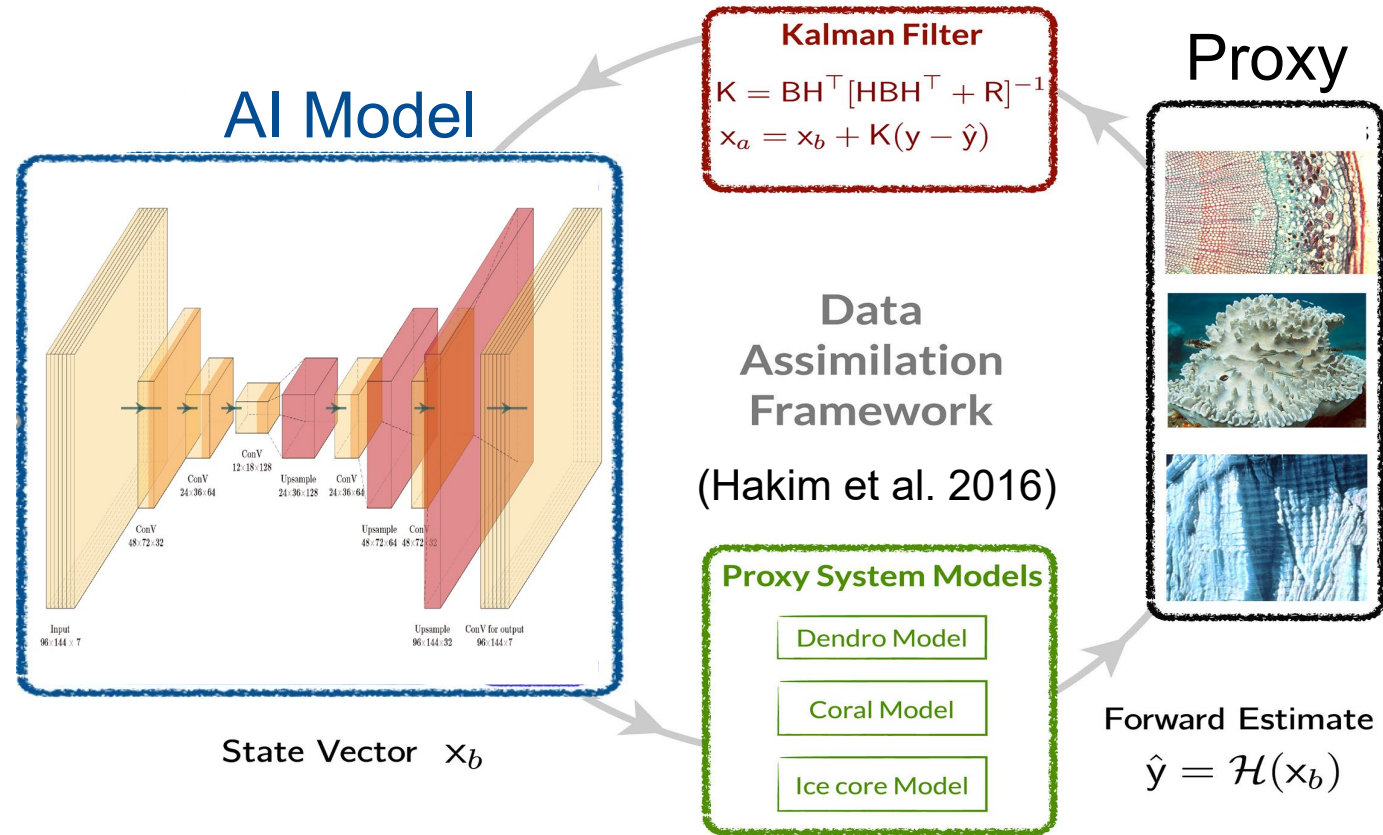
Two goals

1. Enable online PDA
2. Extend online PDA to smoothing

From offline to online PDA

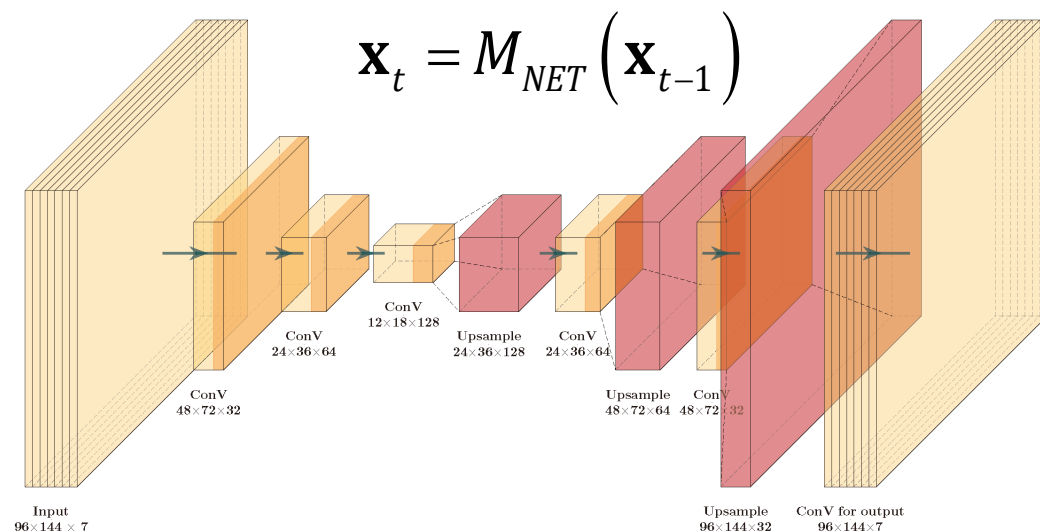


From offline to online PDA



Surrogate models: NET and LIM

CNN-based (NET)



(Scher 2018)

Linear inverse model (LIM)

$$\mathbf{x}_t = \mathbf{M}_{LIM} \mathbf{x}_{t-1}$$

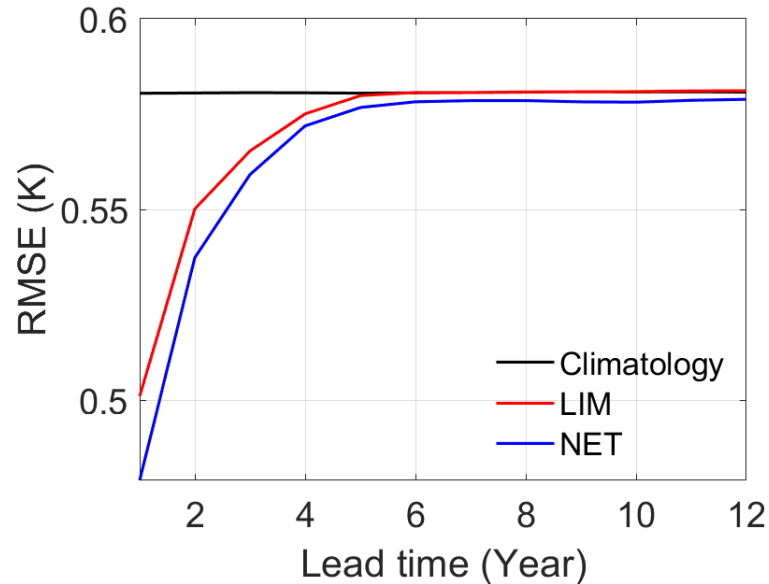
- Linear $\mathbf{M}_{LIM}$ is estimated based on auto-regression
- $\mathbf{M}_{LIM} = \mathbf{C}(1)\mathbf{C}(0)^{-1}$
- $\mathbf{C}(0) = \langle \mathbf{x}_{1:t-1}, \mathbf{x}_{1:t-1}^T \rangle$ $\mathbf{C}(1) = \langle \mathbf{x}_{2:t}, \mathbf{x}_{1:t-1}^T \rangle$

(Perkins and Hakim 2017)

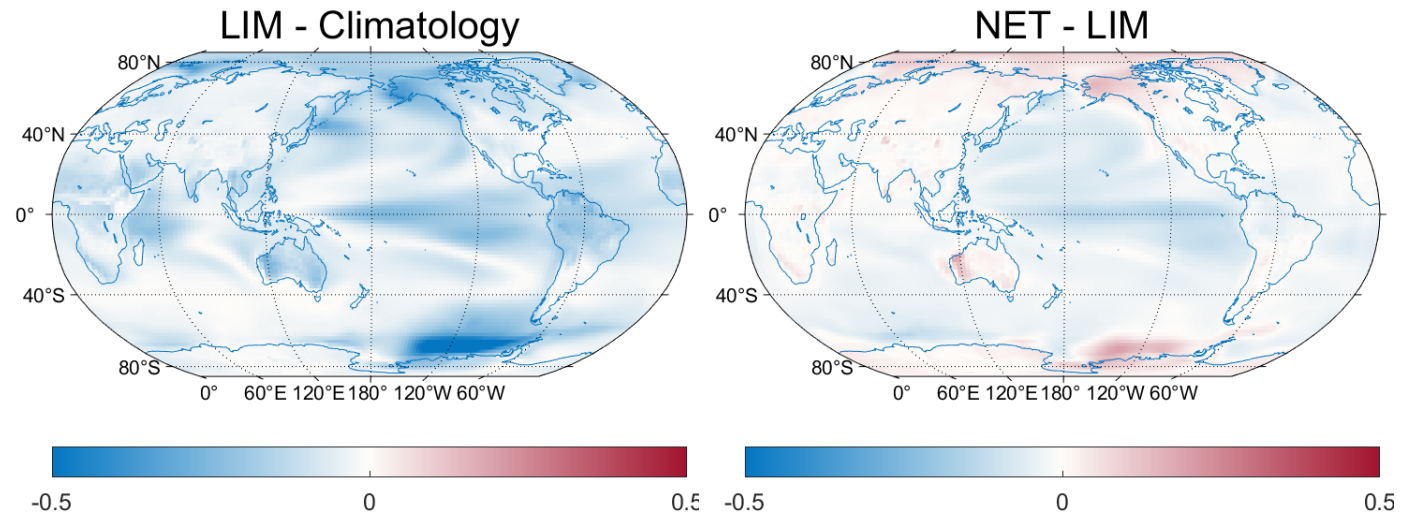
- Propagate temperature from year t to year $t+1$
- Trained on CESM-LME
- LIM: **linear** evolution; NET: **nonlinear** relationships

Surrogate models' predictive skill

Predictive errors

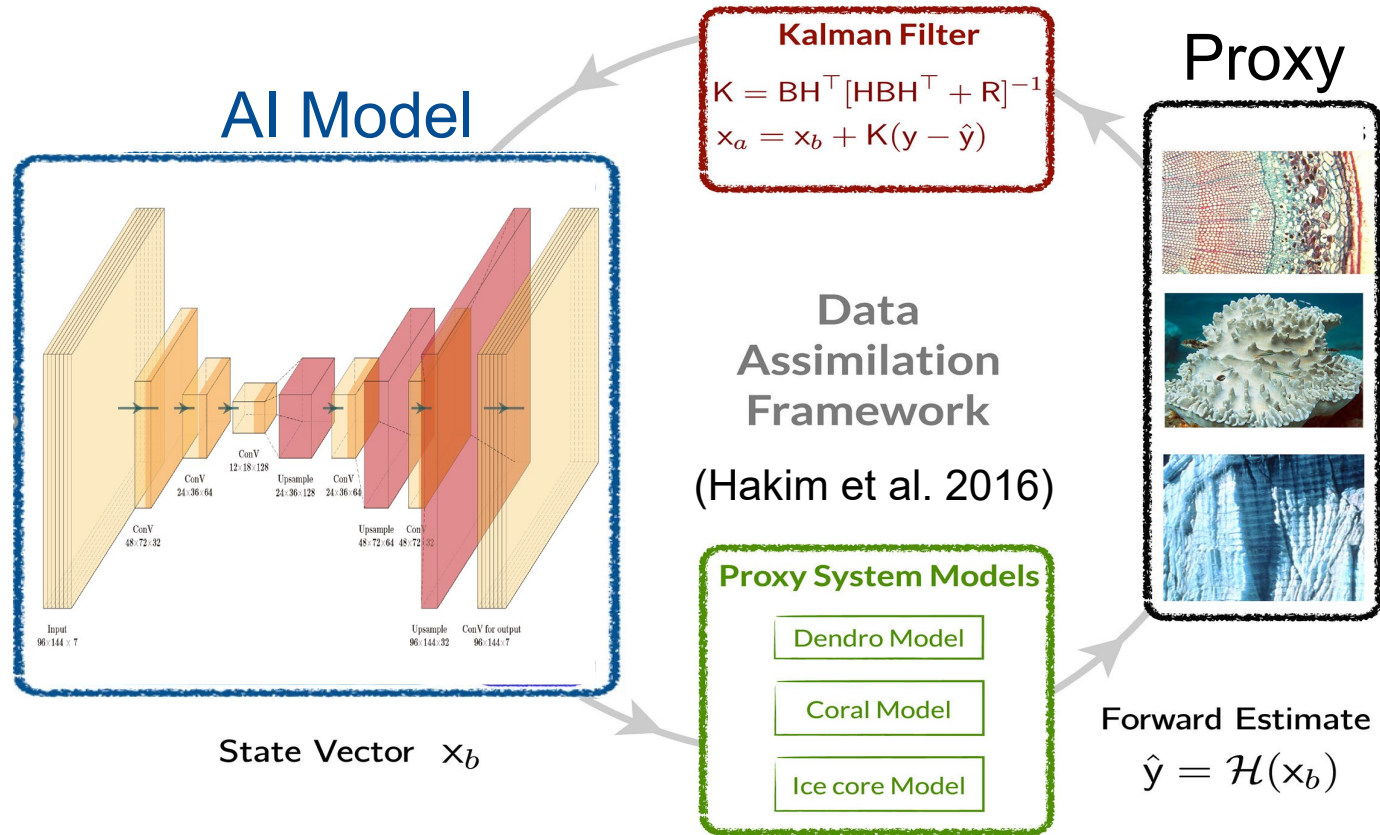


Spatial error reduction

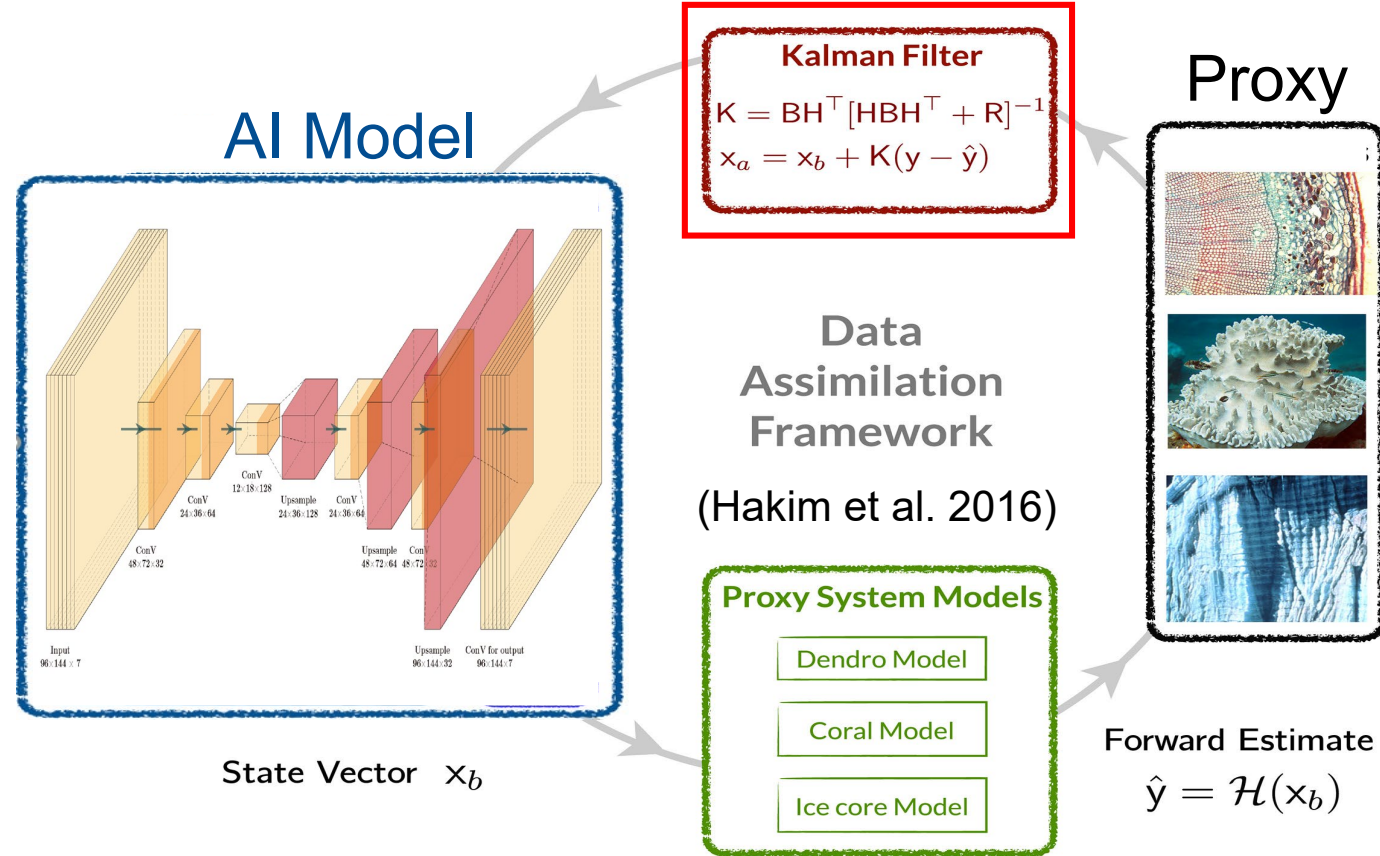


- LIM retains useful predictive skill for ~6 years
- NET further reduces prediction error; improvements mainly locate over the oceans

Coupling AI with PDA



Coupling AI with PDA



AI ensembles introduce new DA challenges

Blend and hybridization

- **Blend:** restore ensemble spread

$$\mathbf{x}_{blend}^f = \beta \mathbf{x}_{AI}^f + (1 - \beta) \mathbf{x}_{climo}^f$$

- **Hybridization:** incorporate climatological error covariance

$$\mathbf{P}_{hyb}^f = \alpha \mathbf{P}_{blend}^f + (1 - \alpha) \mathbf{P}_{climo}^f$$

Blend and hybridization

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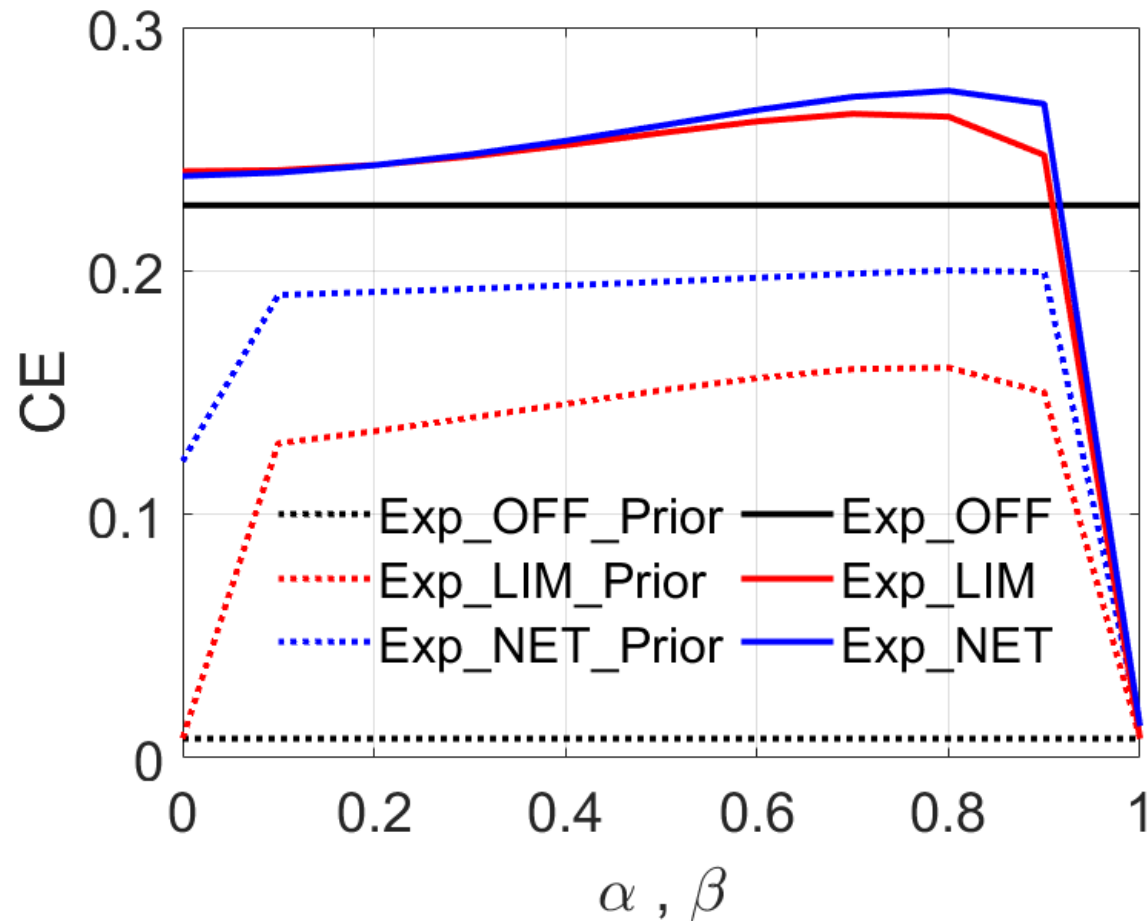
- **Hybridization:** incorporate climatological error covariance

$$\mathbf{P}_{hyb}^f = \alpha \mathbf{P}_{blend}^f + (1 - \alpha) \mathbf{P}_{climo}^f$$

$$\hat{\mathbf{X}}'^f = \left[\frac{\sqrt{\alpha}}{\sqrt{N-1}} \left[\mathbf{x}'_{blend,1}, \mathbf{x}'_{blend,2}, \dots, \mathbf{x}'_{blend,N} \right], \frac{\sqrt{1-\alpha}}{\sqrt{M-1}} \left[\mathbf{x}'_{climo,1}, \mathbf{x}'_{climo,2}, \dots, \mathbf{x}'_{climo,M} \right] \right]$$

$$\mathbf{P}_{hyb}^f = \hat{\mathbf{X}}'^f \left(\hat{\mathbf{X}}'^f \right)^T = \alpha \mathbf{P}_{blend}^f + (1 - \alpha) \mathbf{P}_{climo}^f$$

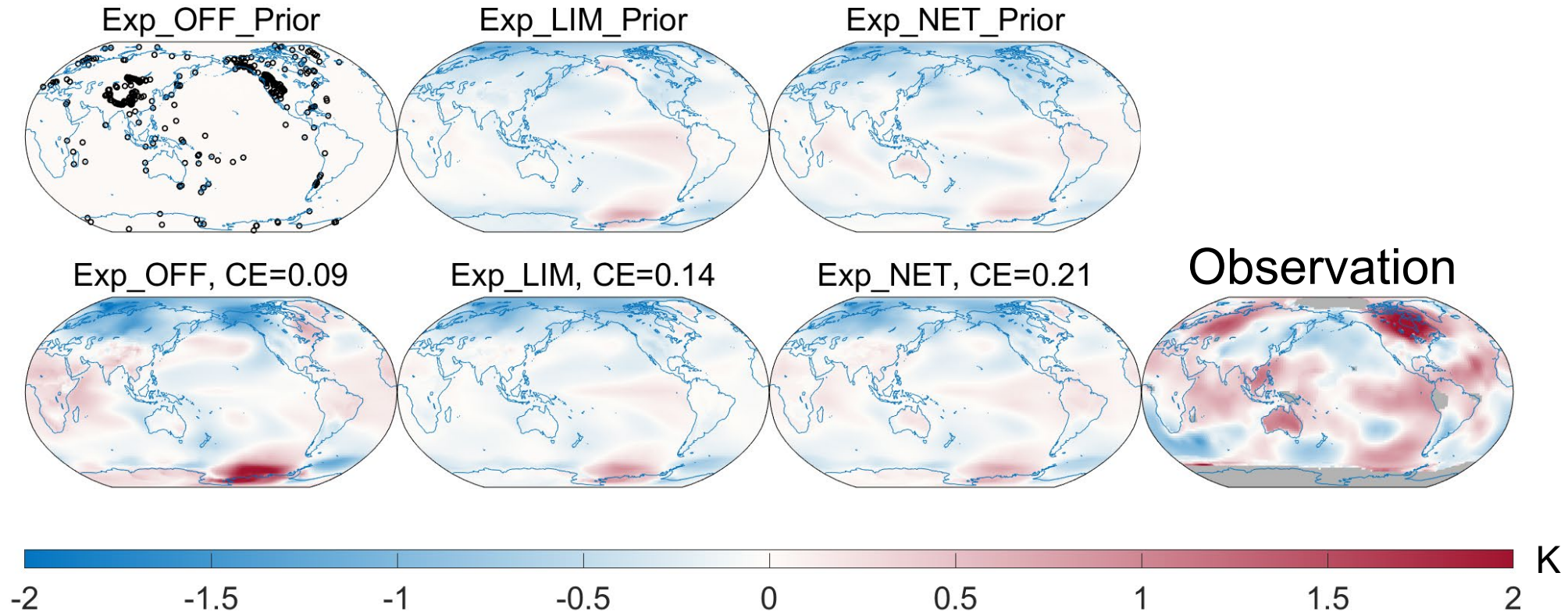
Sensitivity to blend / hybrid weight



- Better priors improve online filtering
- Blend and hybridization stabilize cycling
- AI-based online PDA performs the best

Reconstruction snapshot

El Niño in 1878



- Offline prior has no predictive signal
- Online priors capture the El Niño pattern; NET better represents regional anomalies

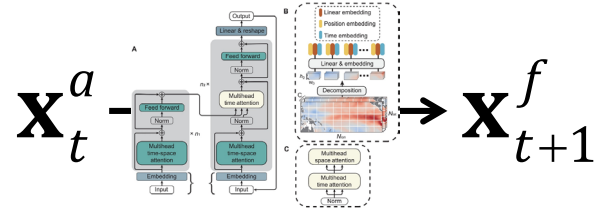
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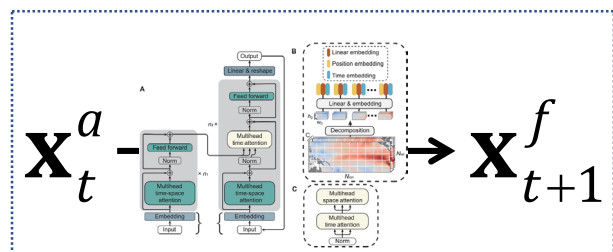
From online filtering to smoothing

$$\mathbf{x}_t^a$$

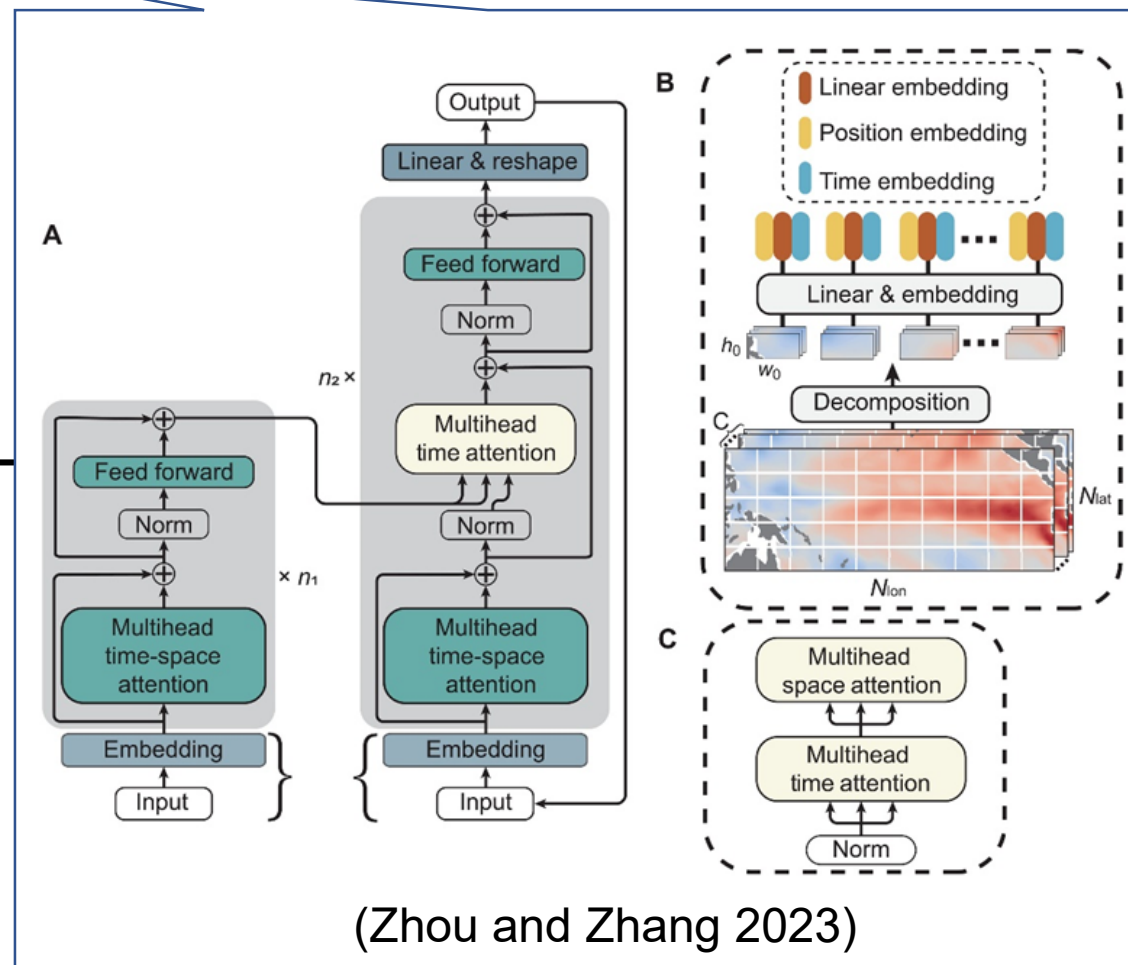
From online filtering to smoothing



From online filtering to smoothing



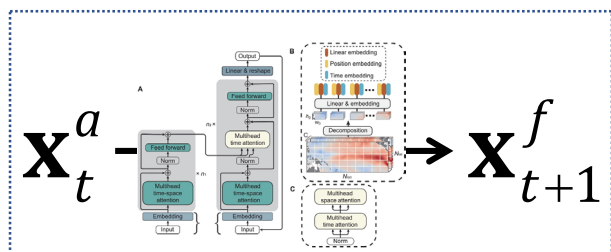
$\mathbf{x}_t^a[1,2,\dots,12]$



(Zhou and Zhang 2023)

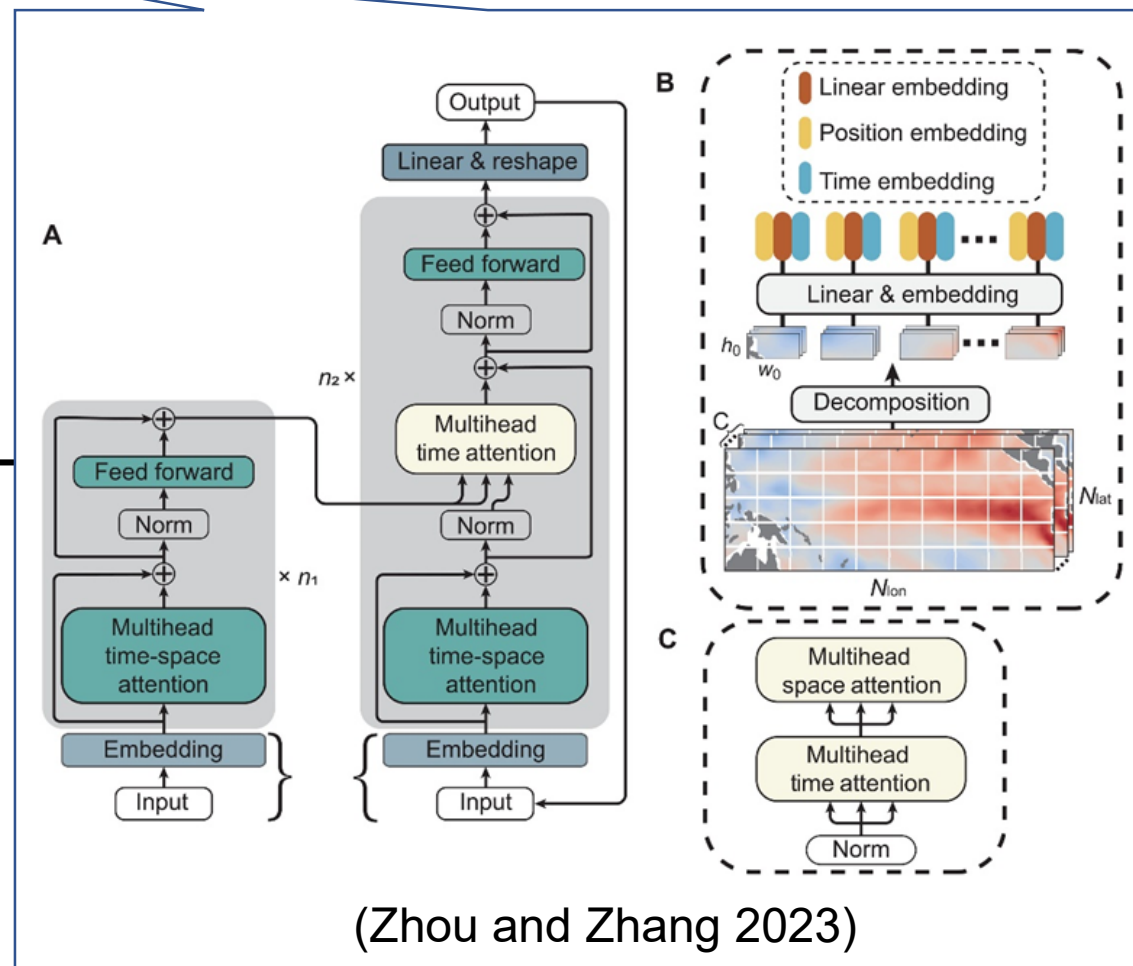
$\mathbf{x}_{t+1}^f[1,2,\dots,12]$

From online filtering to smoothing



$$\mathbf{x}_t^a[1,2,\dots,12]$$

$$\begin{bmatrix} \text{SAT}_t \\ \text{SST}_t \end{bmatrix}$$



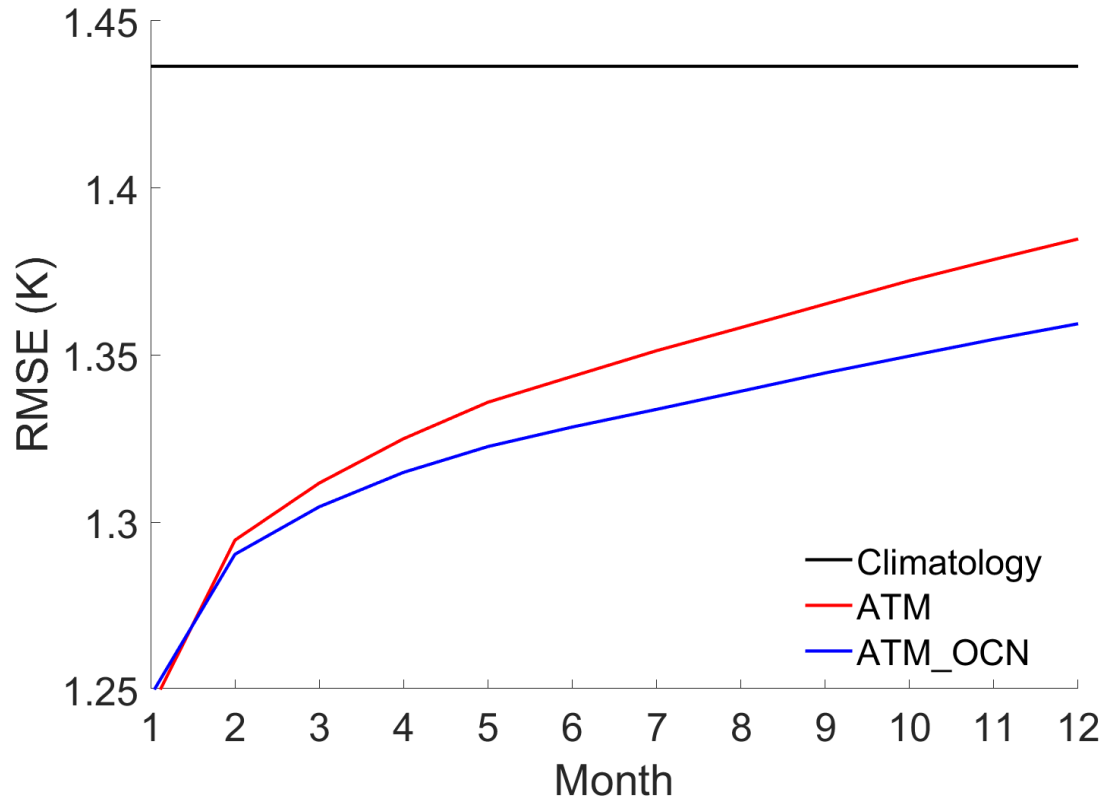
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$$\begin{bmatrix} \text{SAT}_{t+1} \\ \text{SST}_{t+1} \end{bmatrix}$$

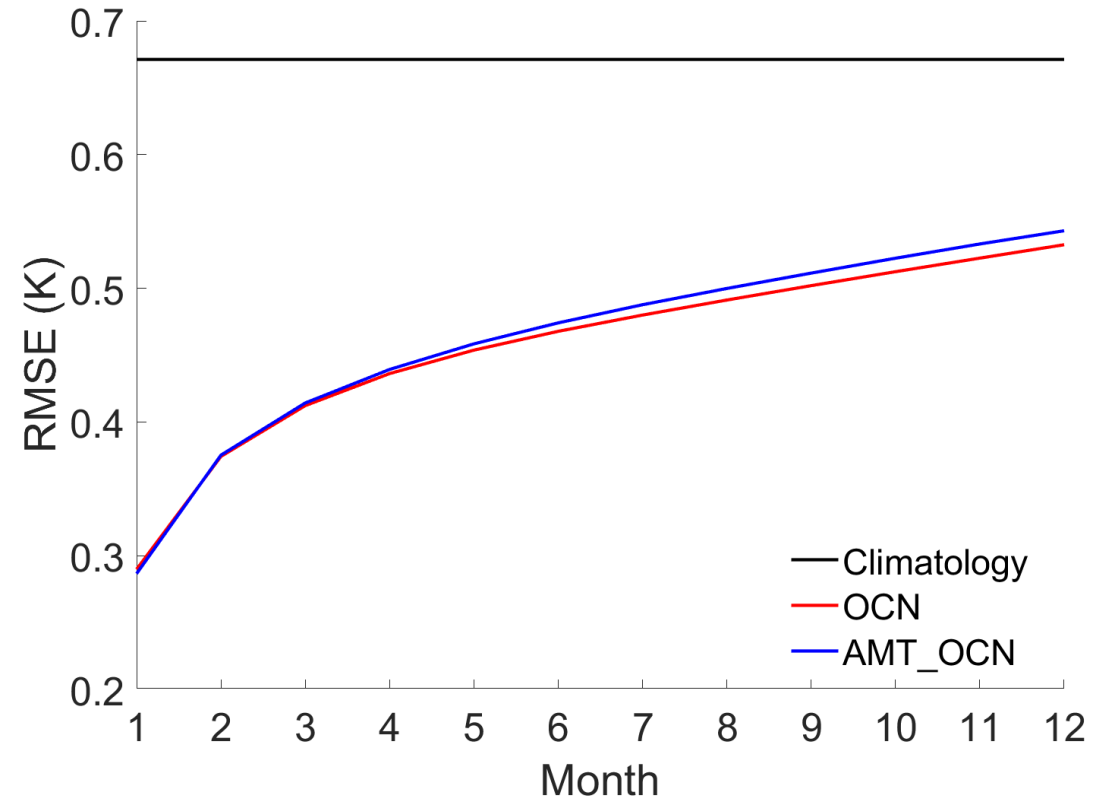
(Zhou and Zhang 2023)

Atmosphere-ocean coupled model

SAT forecast



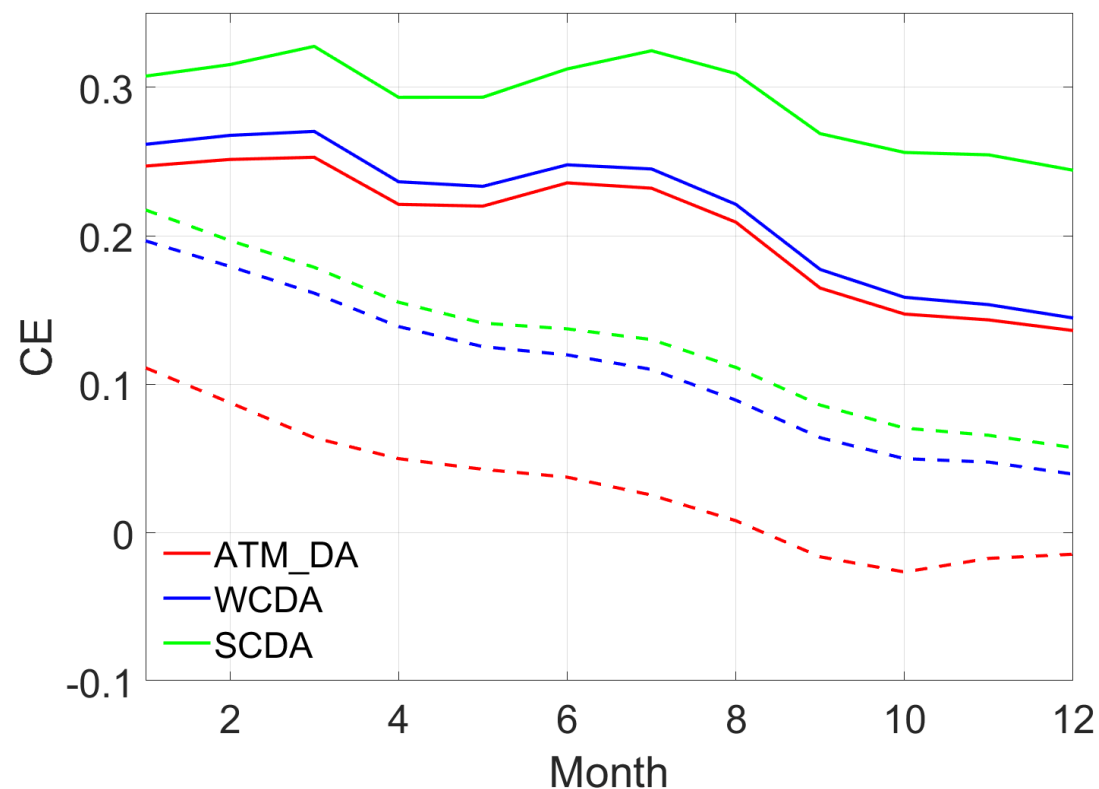
SST forecast



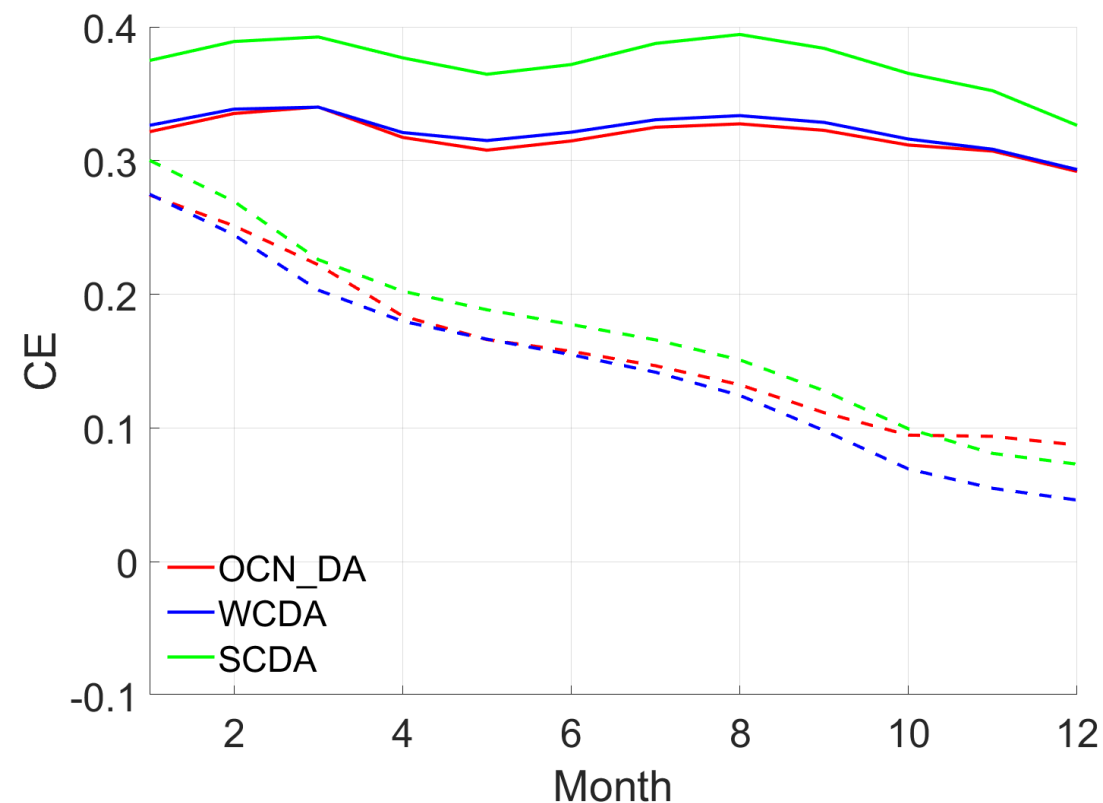
- Coupling SST improves SAT forecasts
- Coupling SAT has limited impact on SST forecasts

Strong coupled DA

SAT reconstruction



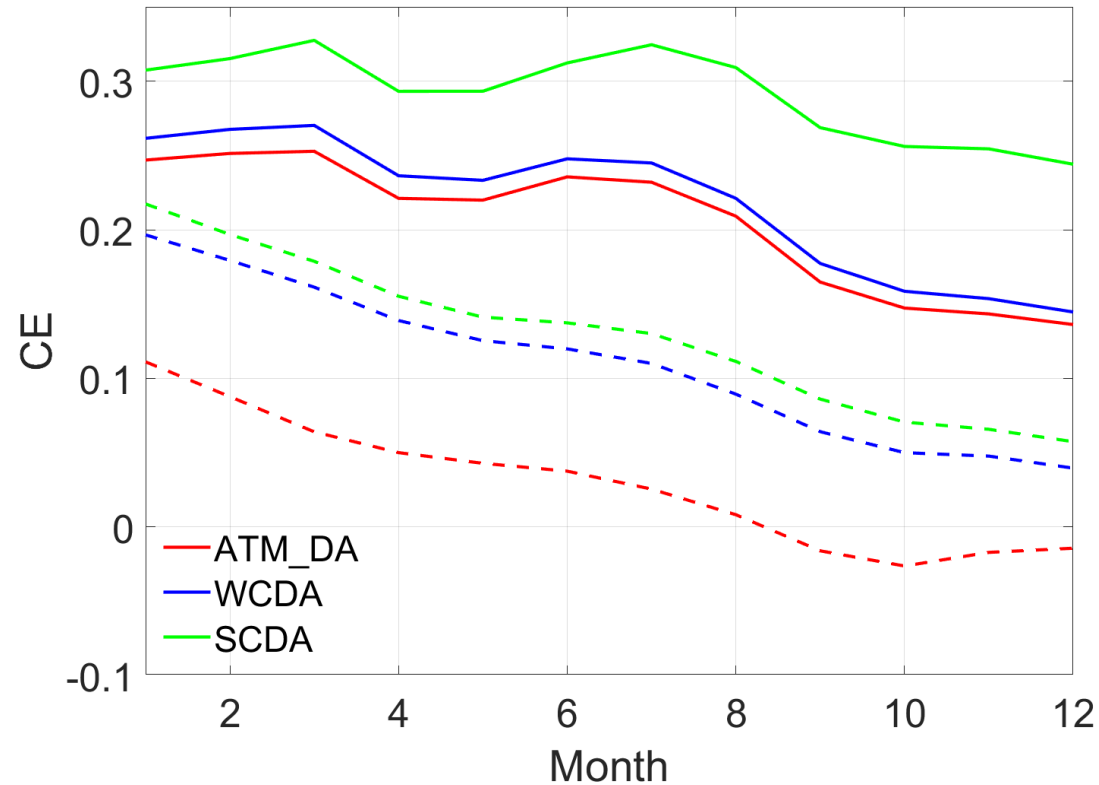
SST reconstruction



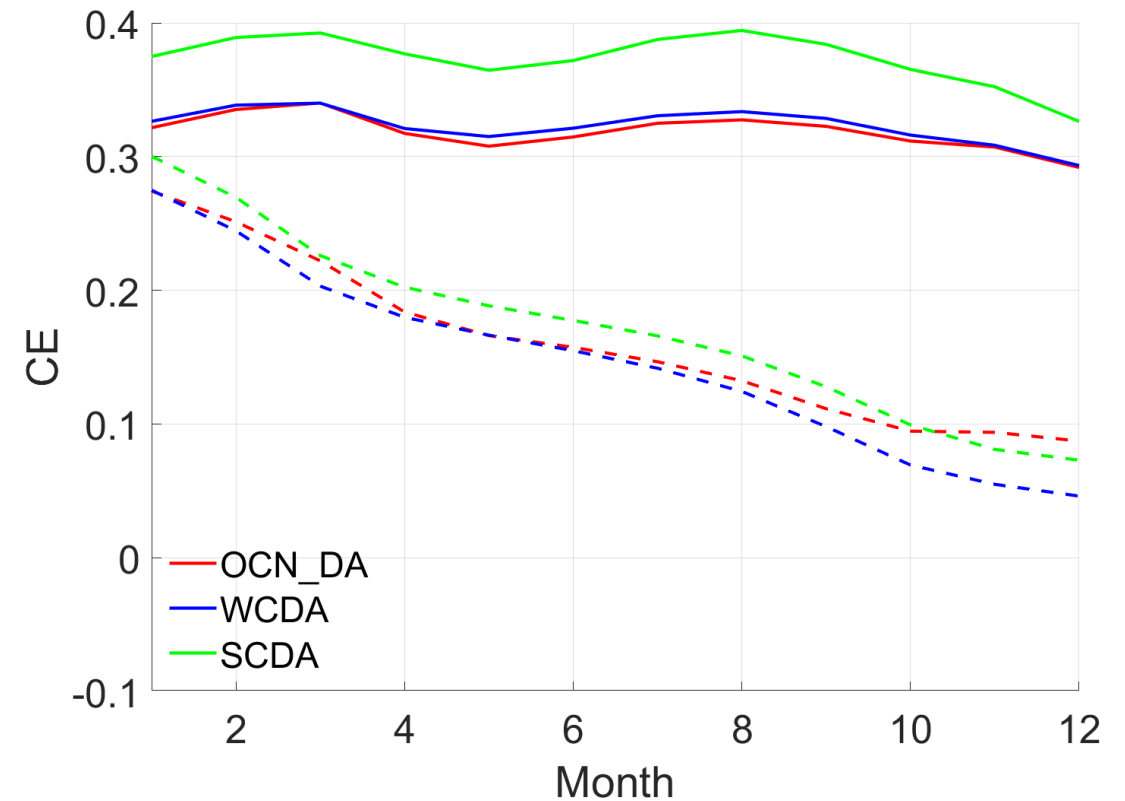
- Coupled model improves the atmospheric prior
- Atmosphere-ocean cross-updates are essential for both posteriors

Strong coupled DA

SAT reconstruction

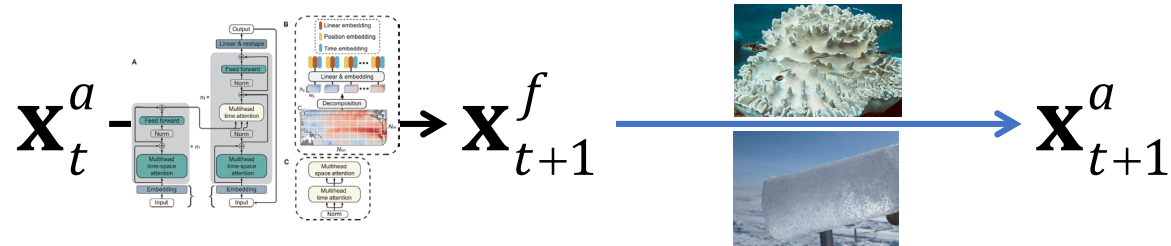


SST reconstruction

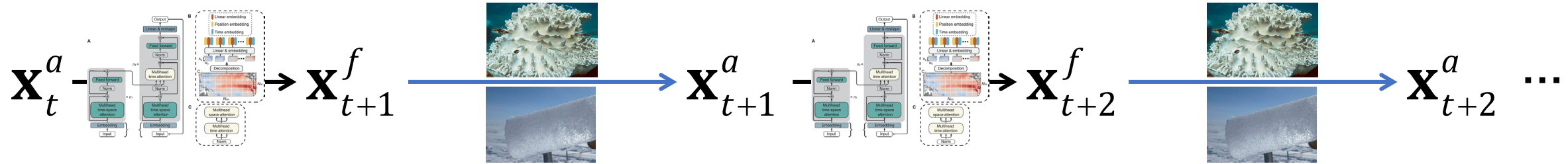


L. Lei, H. Sun, Z. Tan, L. Ning, and Z. Liu, 2026: Strongly Coupled Data Assimilation for Paleoclimate Reconstruction Using Deep Learning-Based Models. [*J. Adv. Model. Earth Syst.*](#)

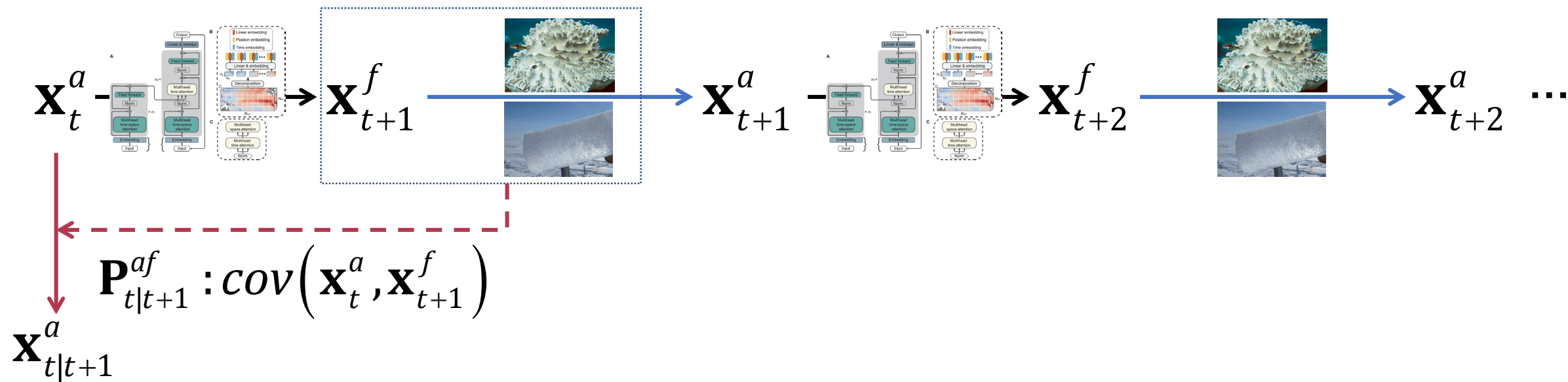
From online filtering to smoothing



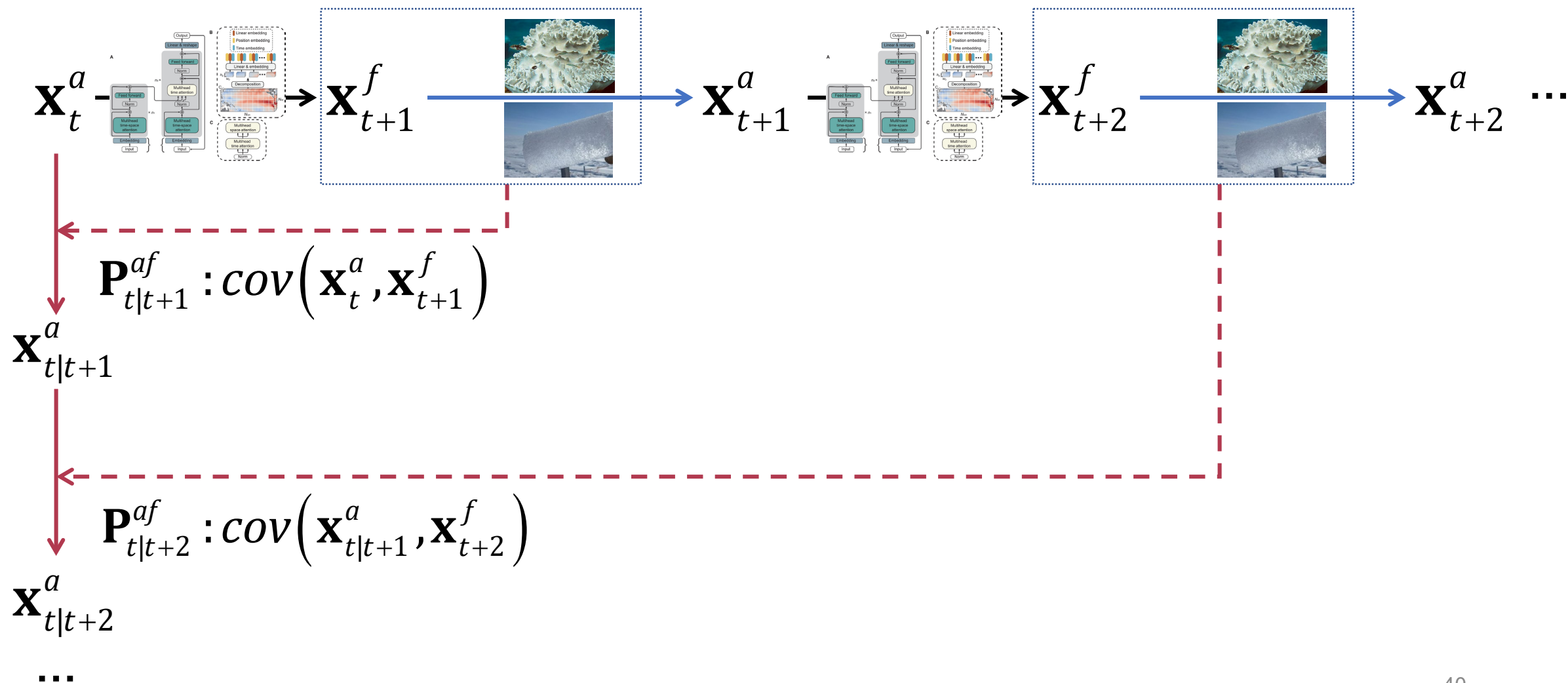
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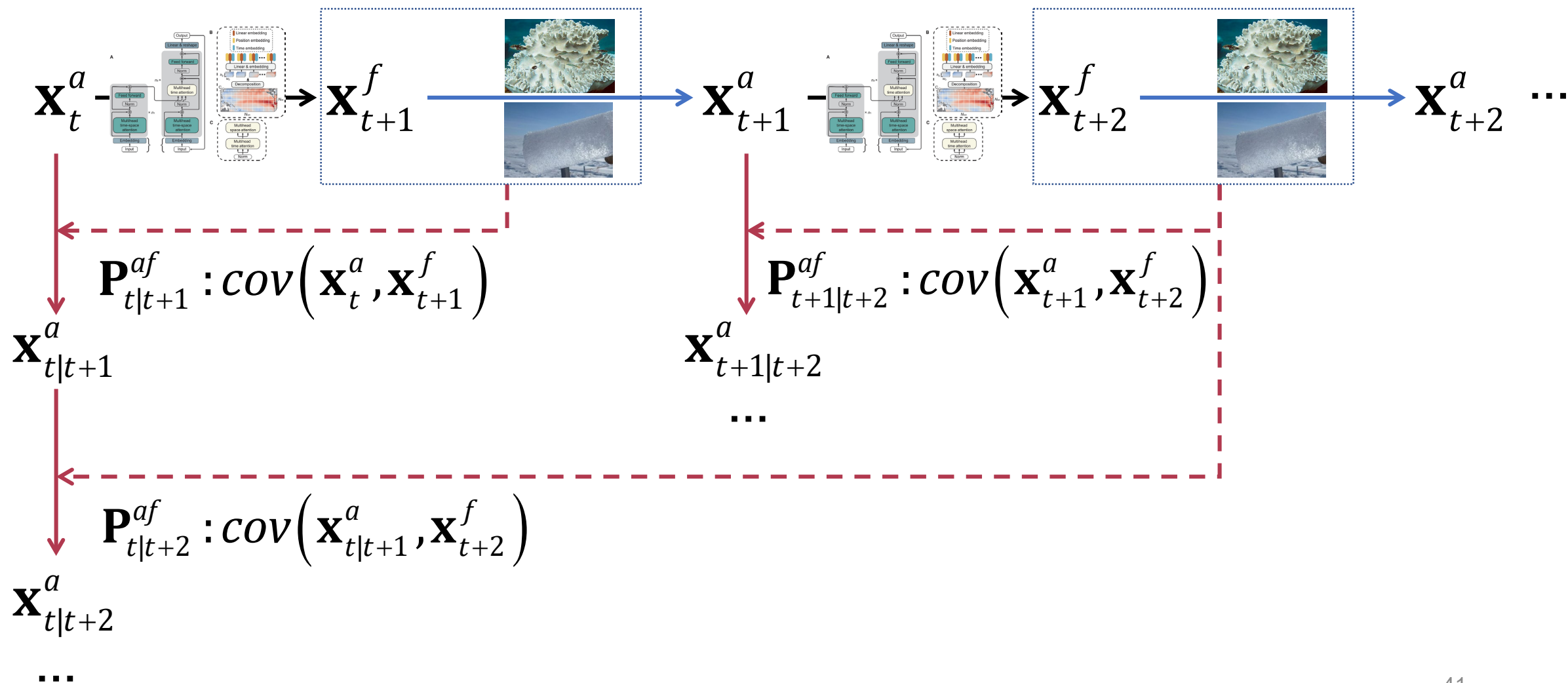
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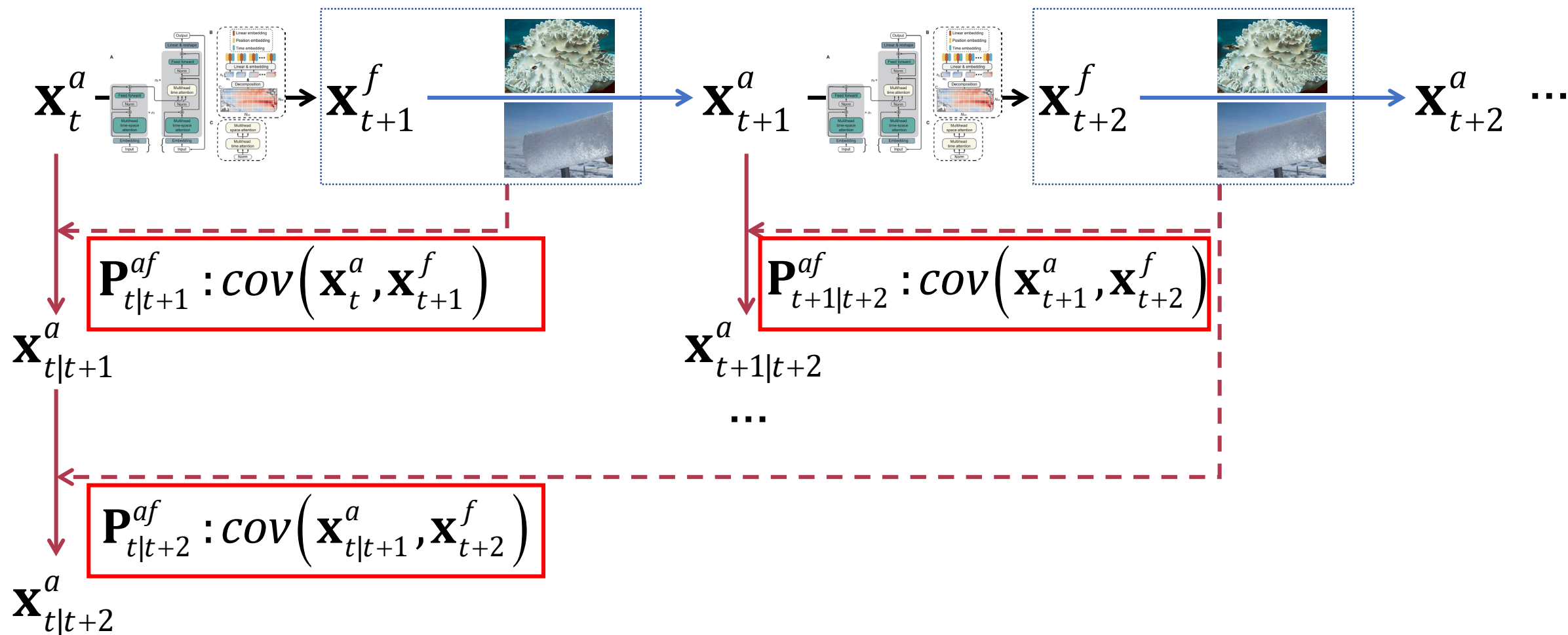
From online filtering to smoothing



From online filtering to smoothing



From online filtering to smoothing



... Cross-time covariances suffer from large sampling errors

Localization and hybridization for smoothing

- **Localization:** suppress spurious cross-time correlations

$$\rho_{t|t+l} \circ \mathbf{P}_{t|t+l}^{af}$$

- **Hybridization:** incorporate climatological cross-time covariance

$$\mathbf{P}_{hyb,t|t+l}^{af} = \gamma \mathbf{P}_{t|t+l}^{af} + (1 - \gamma) \mathbf{P}_{climo,t|t+l}^f$$

Localization and hybridization for smoothing

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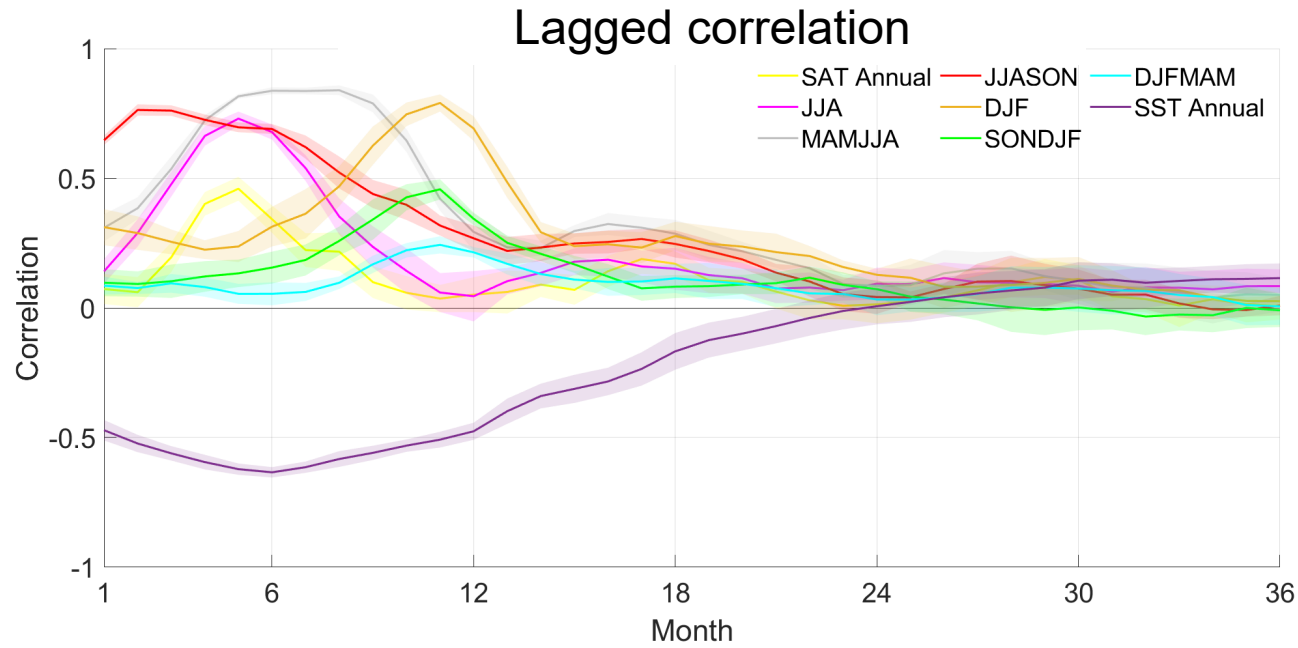
$$\mathbf{P}_{hyb,t|t+l}^{af} = \gamma \mathbf{P}_{t|t+l}^{af} + (1 - \gamma) \mathbf{P}_{climo,t|t+l}^f$$

$$\hat{\mathbf{X}}_t'^a = \left[\frac{\sqrt{\gamma}}{\sqrt{N-1}} [\mathbf{x}_{1,t}'^a, \mathbf{x}_{2,t}'^a, \dots, \mathbf{x}_{N,t}'^a], \frac{\sqrt{1-\gamma}}{\sqrt{M-1}} [\mathbf{x}_{climo,1,t}'^f, \mathbf{x}_{climo,2,t}'^f, \dots, \mathbf{x}_{climo,M,t}'^f] \right]$$

$$\hat{\mathbf{X}}_{t+l}'^f = \left[\frac{\sqrt{\gamma}}{\sqrt{N-1}} [\mathbf{x}_{1,t+l}'^f, \mathbf{x}_{2,t+l}'^f, \dots, \mathbf{x}_{N,t+l}'^f], \frac{\sqrt{1-\gamma}}{\sqrt{M-1}} [\mathbf{x}_{climo,1,t+l}'^f, \mathbf{x}_{climo,2,t+l}'^f, \dots, \mathbf{x}_{climo,M,t+l}'^f] \right]$$

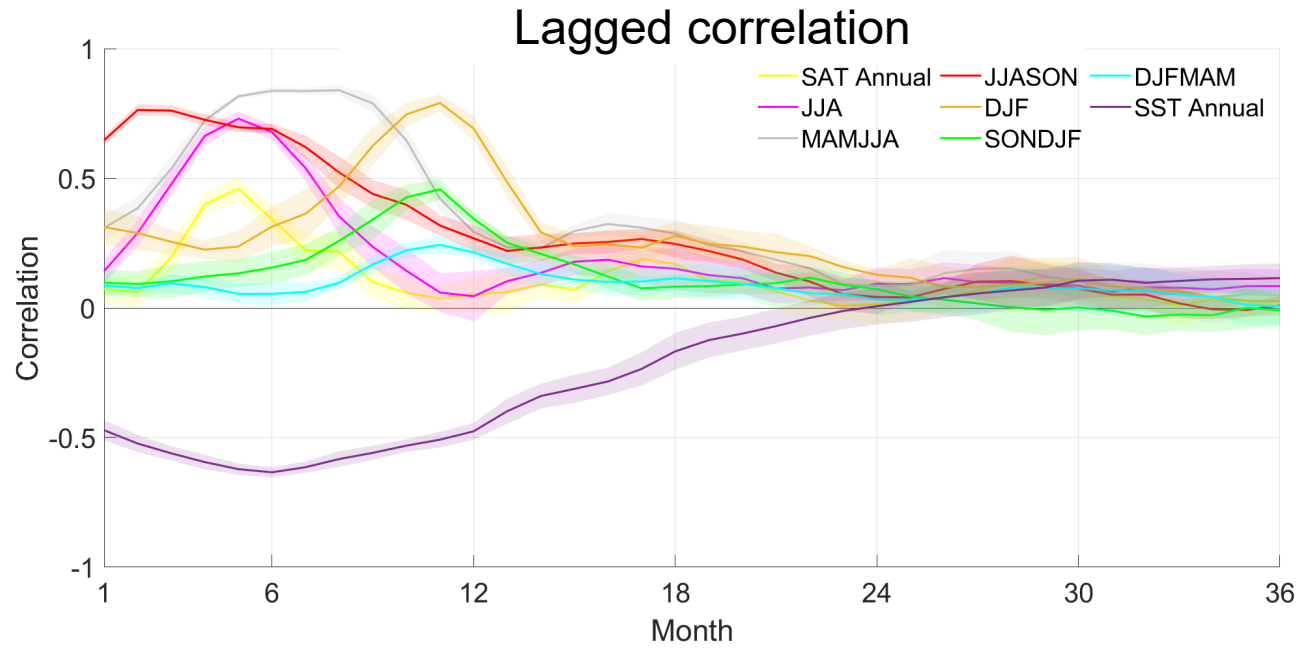
$$\mathbf{P}_{hyb,t|t+l}^{af} = \hat{\mathbf{X}}_t'^a \left(\hat{\mathbf{X}}_{t+l}'^f \right)^T = \gamma \mathbf{P}_{t|t+l}^{af} + (1 - \gamma) \mathbf{P}_{climo,t|t+l}^f$$

Adaptive temporal localization

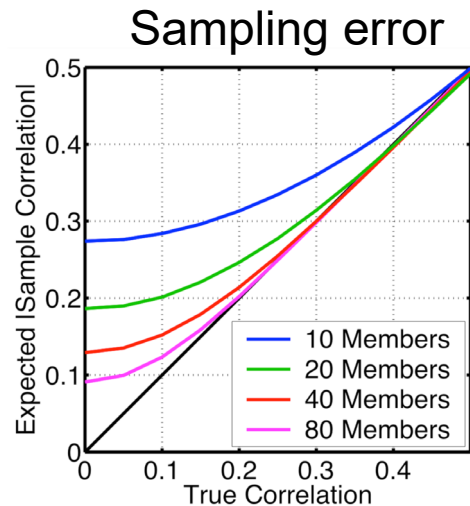


- Correlation weakens with increasing lag time

Adaptive temporal localization

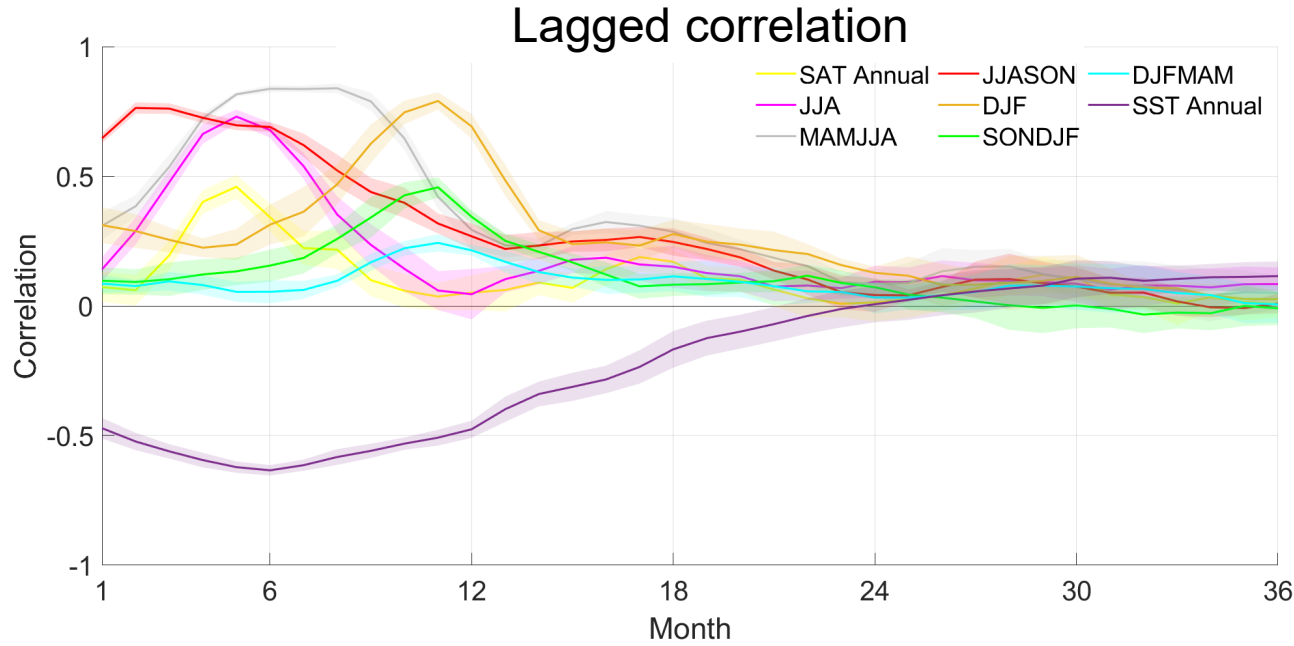


- Correlation weakens with increasing lag time
- Sampling error increases as correlation weakens

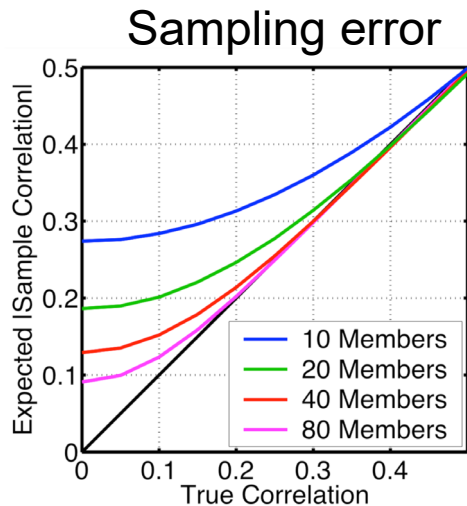


(DART tutorial)

Adaptive temporal localization



- Correlation weakens with increasing lag time
- Sampling error increases as correlation weakens
- GGF accounts for uncertainty in the estimated correlations



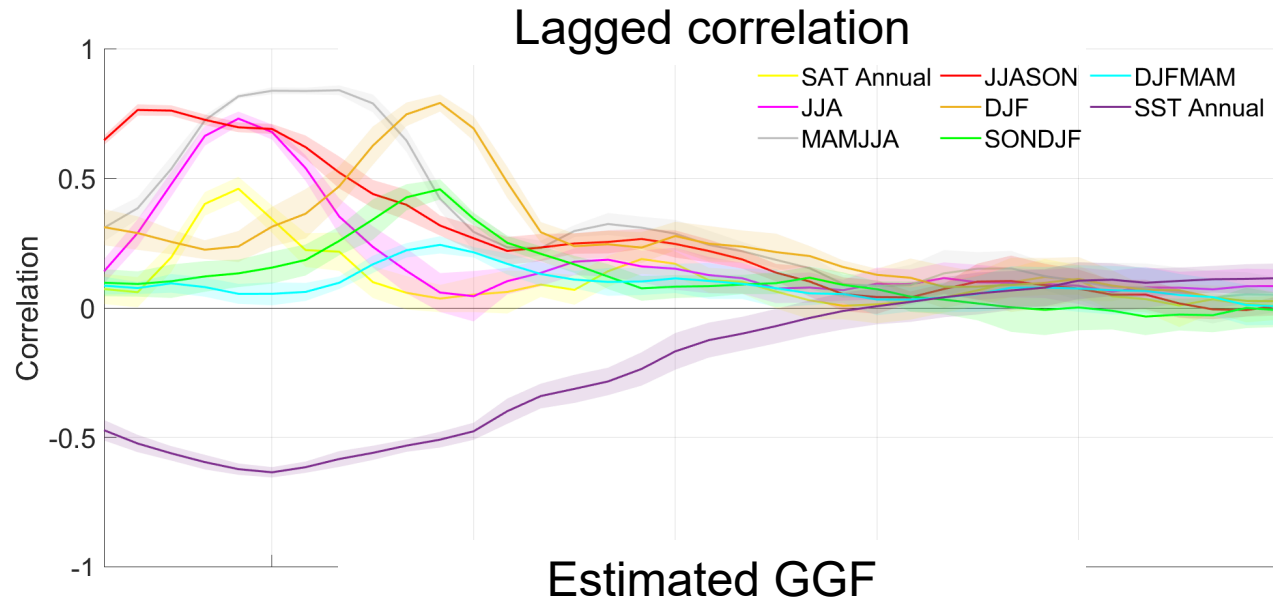
(DART tutorial)

Global Group Function (GGF)

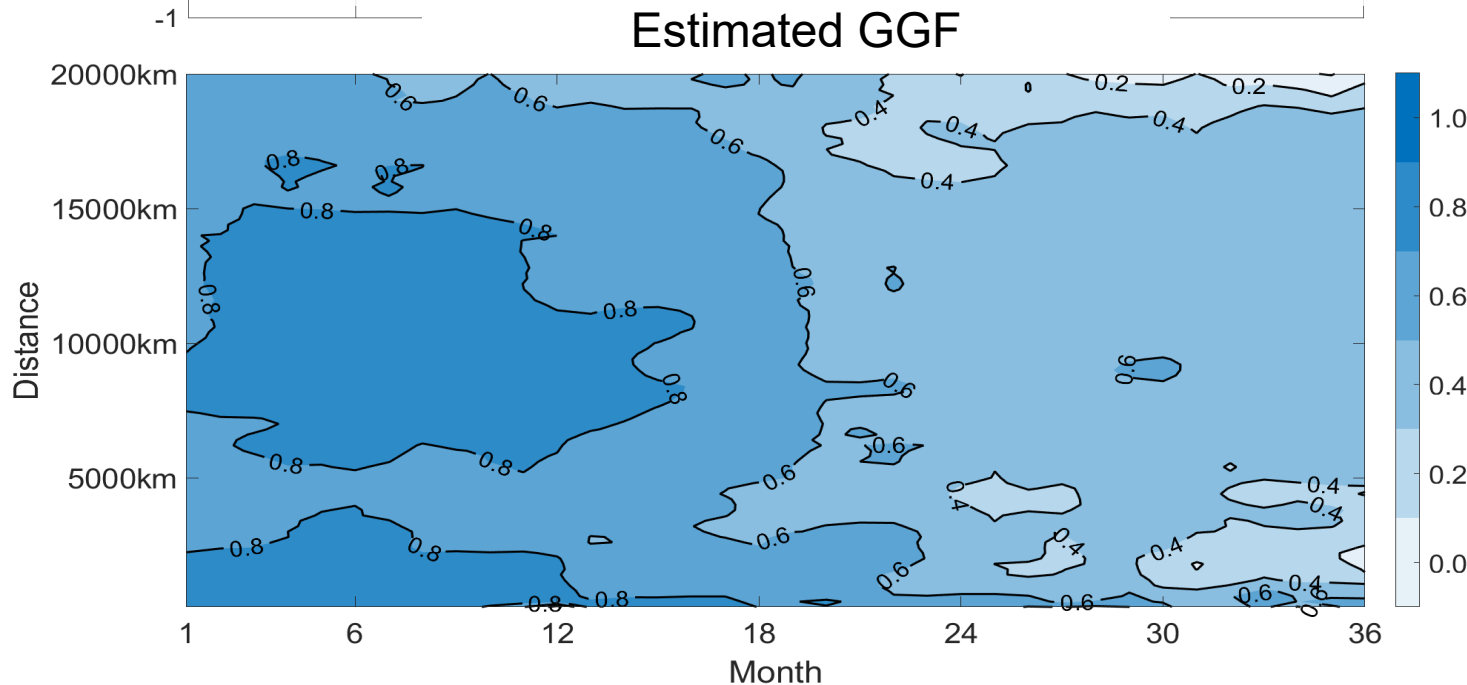
$$J = \sqrt{\sum_{i=1}^G \sum_{j=1}^G (\rho \hat{r}_j - \hat{r}_i)^2}$$
$$\rho^* = \underset{\rho}{\operatorname{argmin}}(J)$$
$$= \left\{ \left[\left(\sum_{i=1}^G \hat{r}_i \right)^2 / \sum_{i=1}^G (\hat{r}_i)^2 \right] - 1 \right\} / (G - 1)$$

(Anderson 2007; Lei et al. 2016)

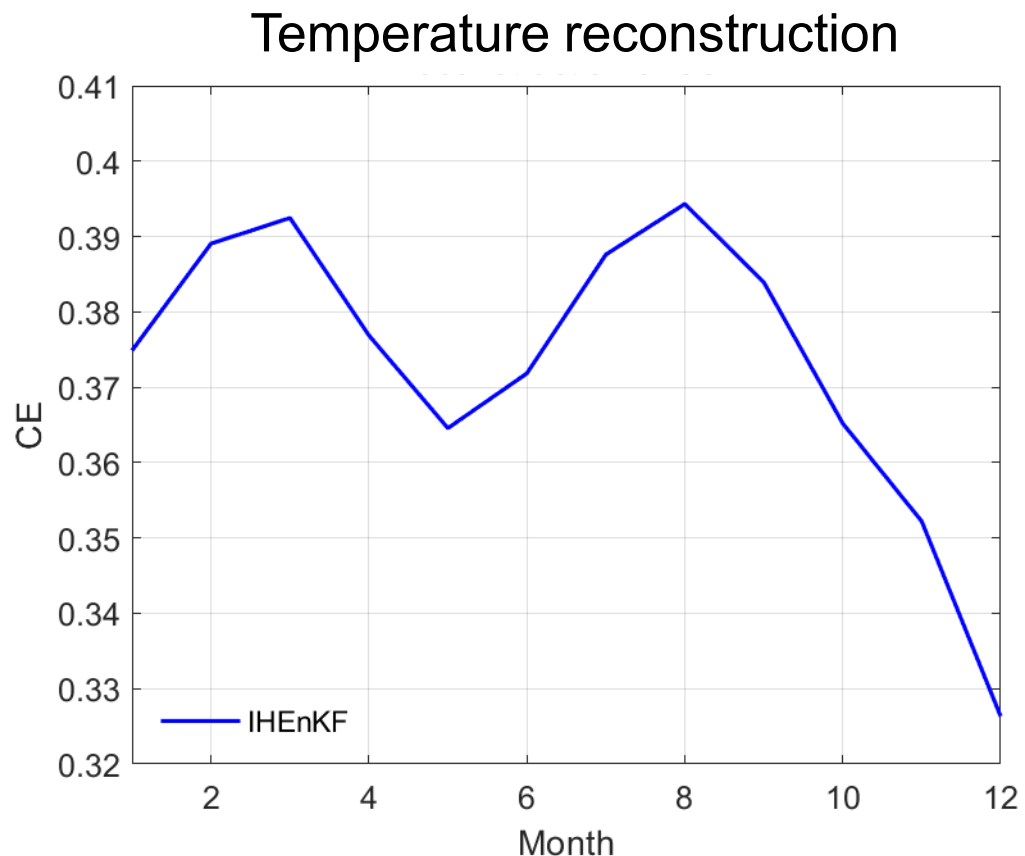
Adaptive temporal localization



- Correlation weakens with increasing lag time
- Sampling error increases as correlation weakens
- GGF accounts for uncertainty in the estimated correlations
- GGF localization decreases with increasing lag time



Reconstruction skill evaluation

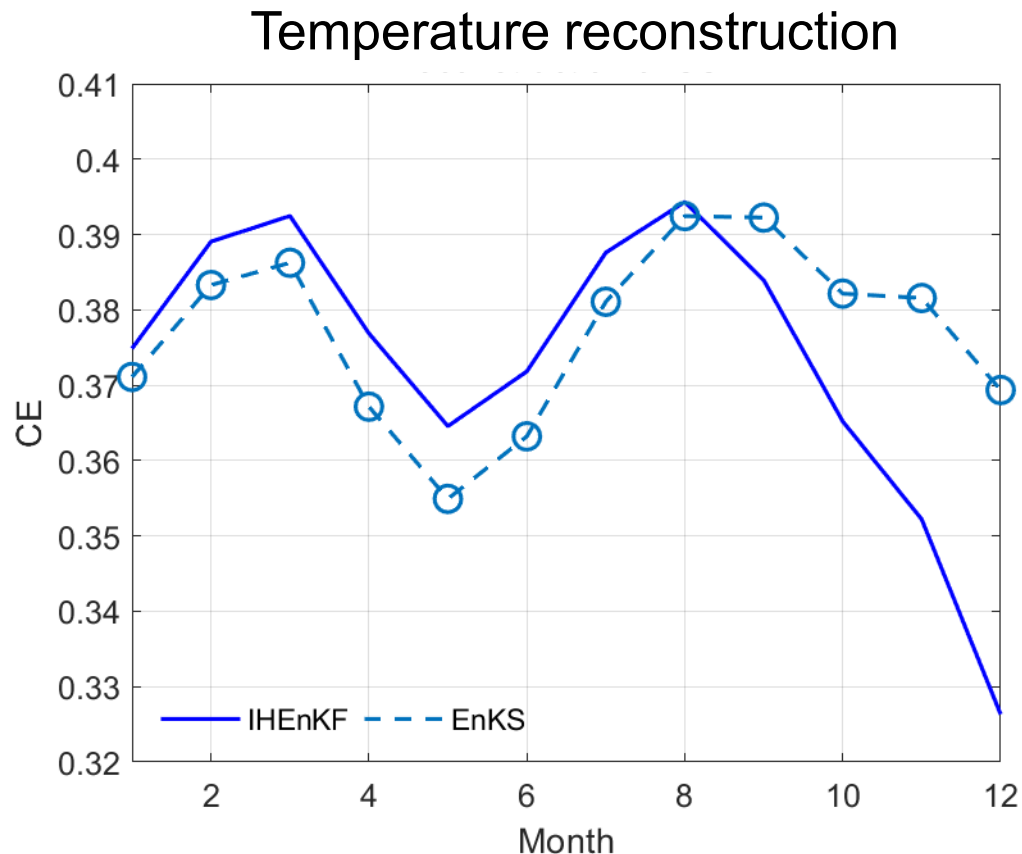


- The last 4 months' skill is low due to the model's limited predictability

Shorter lead time → Longer lead time

(Longer lag time) → (Shorter lag time)

Reconstruction skill evaluation

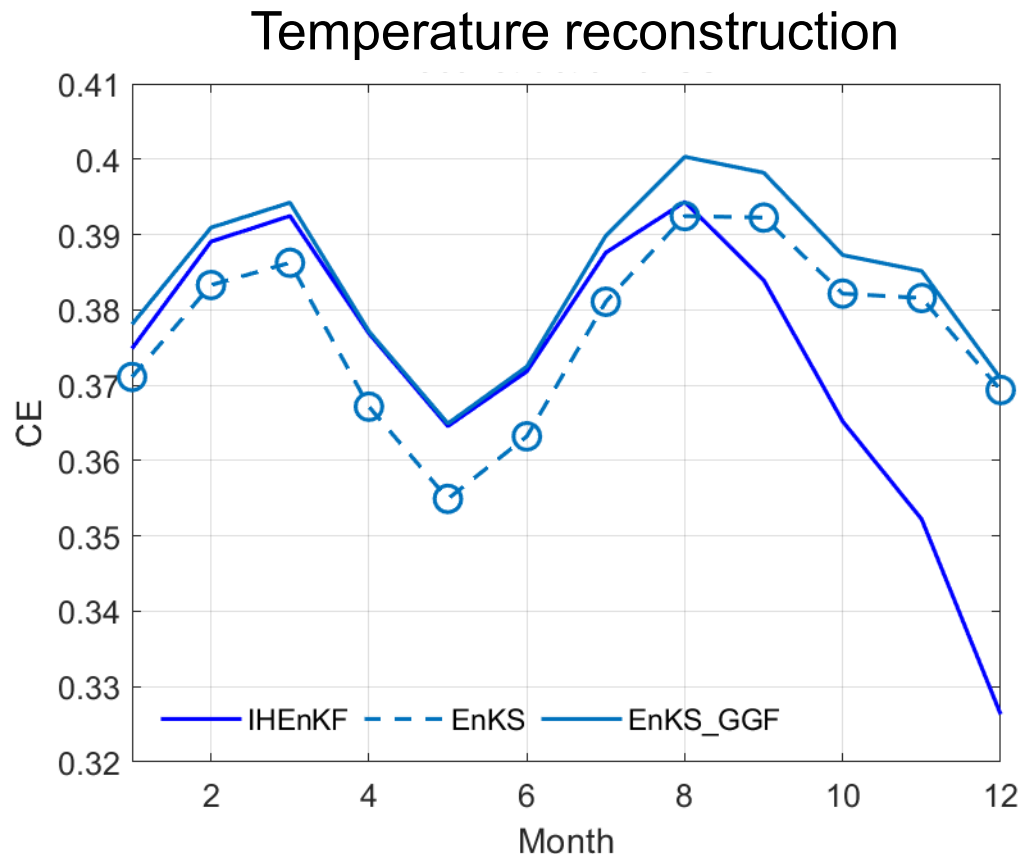


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- EnKS improves the last 4 months but degrades the first 8 months

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Reconstruction skill evaluation

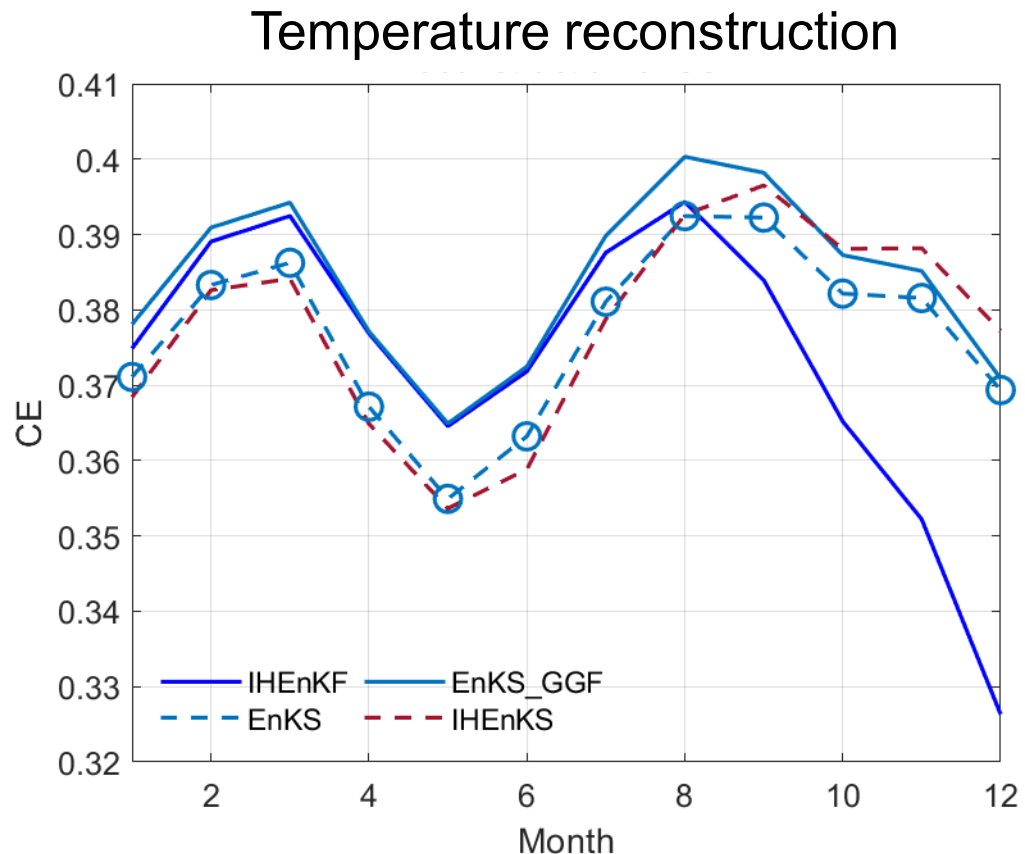


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- EnKS_GGF improves skill across all 12 months, particularly in the first 8 months

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Reconstruction skill evaluation

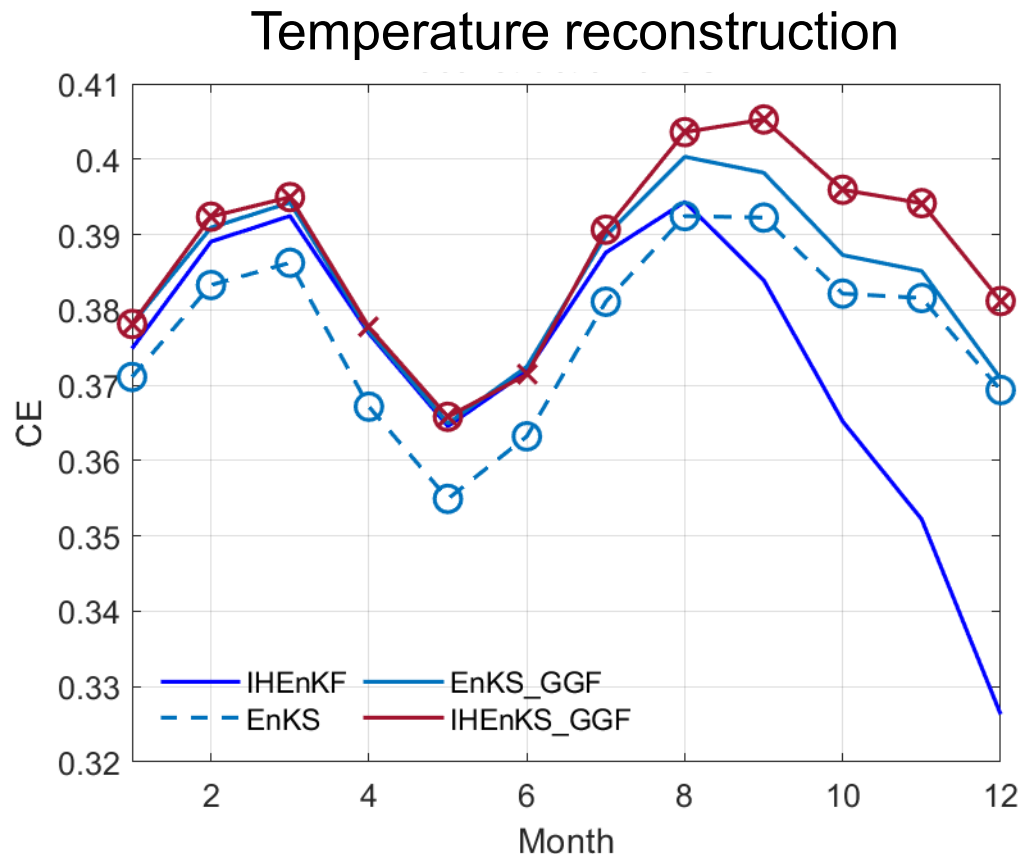


Shorter lead time → **Longer lead time**

(Longer lag time) → **(Shorter lag time)**

- The last 4 months' skill is low due to the model's limited predictability
- EnKS improves the last 4 months but degrades the first 8 months
- EnKS_GGF improves skill across all 12 months, particularly in the first 8 months
- IHEEnKS improves the last 4 months but degrades the first 8 months

Reconstruction skill evaluation



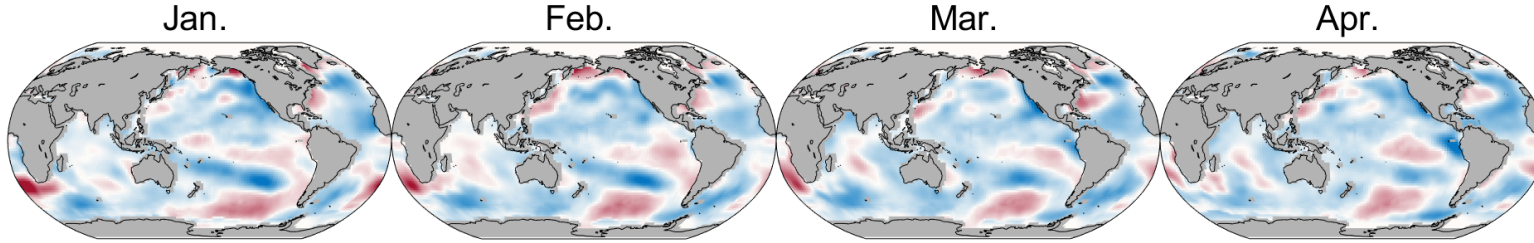
- The last 4 months' skill is low due to the model's limited predictability
- EnKS improves the last 4 months but degrades the first 8 months
- EnKS_GGF improves skill across all 12 months, particularly in the first 8 months
- IHEEnKS_GGF further improves skill in the last 4 months

Shorter lead time → Longer lead time

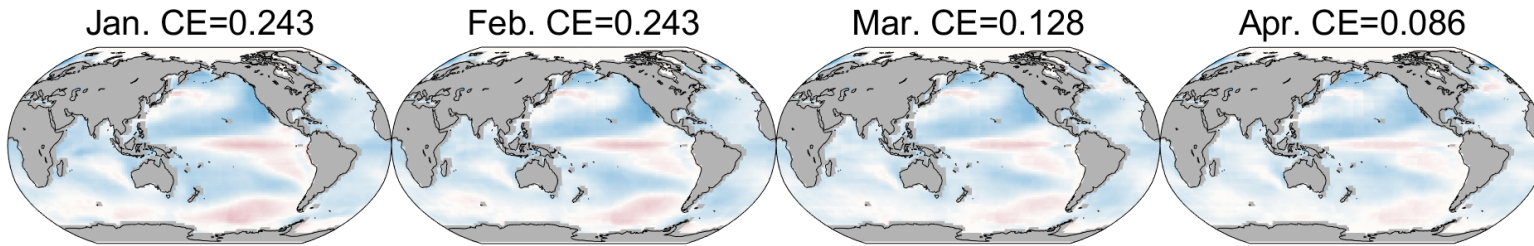
(Longer lag time) → (Shorter lag time)

Reconstruction snapshot

1913 ERSST



1913 IEnKF

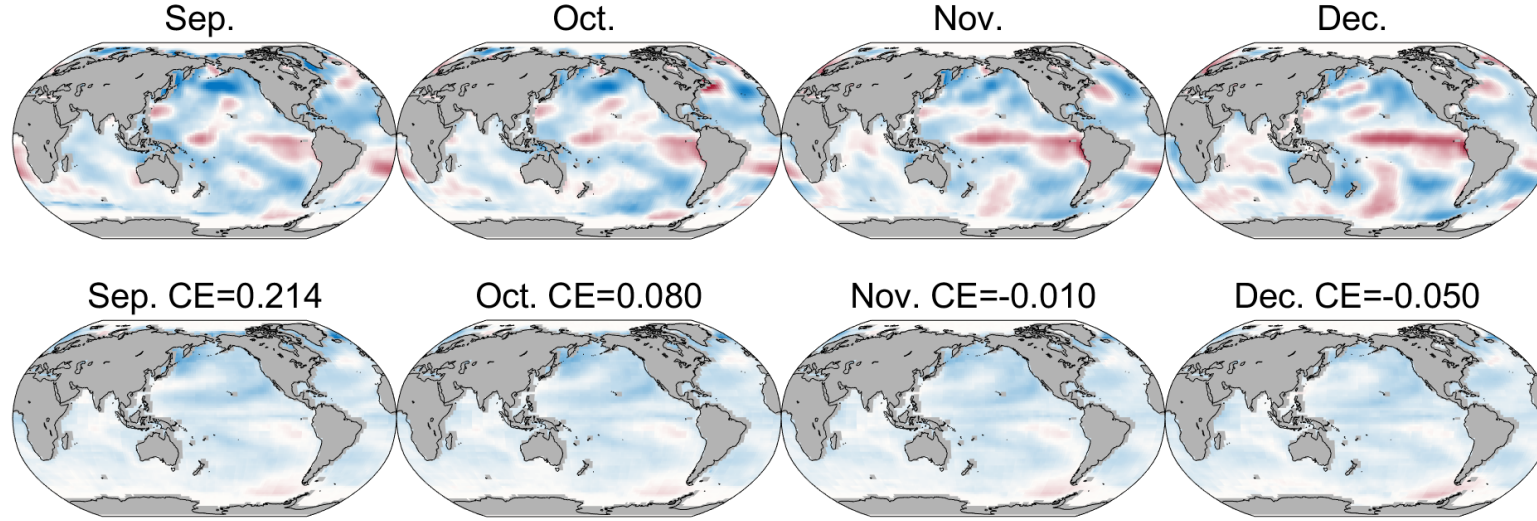


- Online filter works well for the first 4 months as lead time is small

Reconstruction snapshot

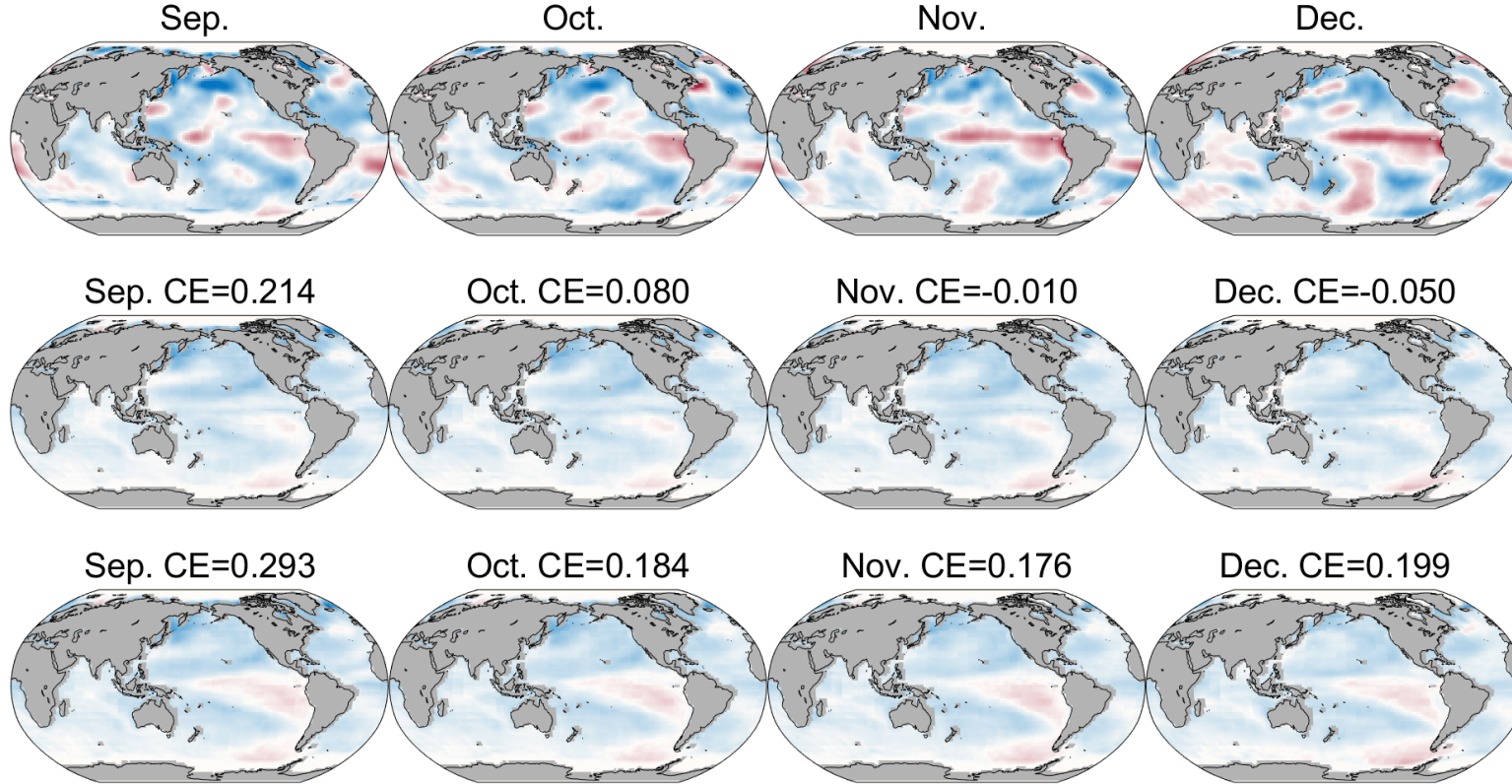
1913 ERSST

1913 IEnKF



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- Limited predictability prevents the online filter from capturing the intensification of El Niño

Reconstruction snapshot



- Online filter works well for the first 4 months as lead time is small
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- Online smoother reconstructs El Niño by utilizing future proxies

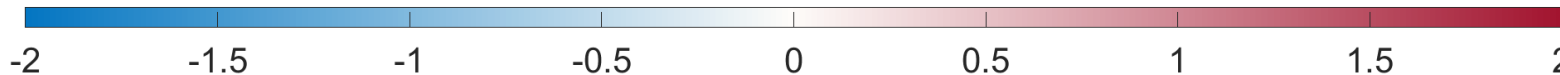
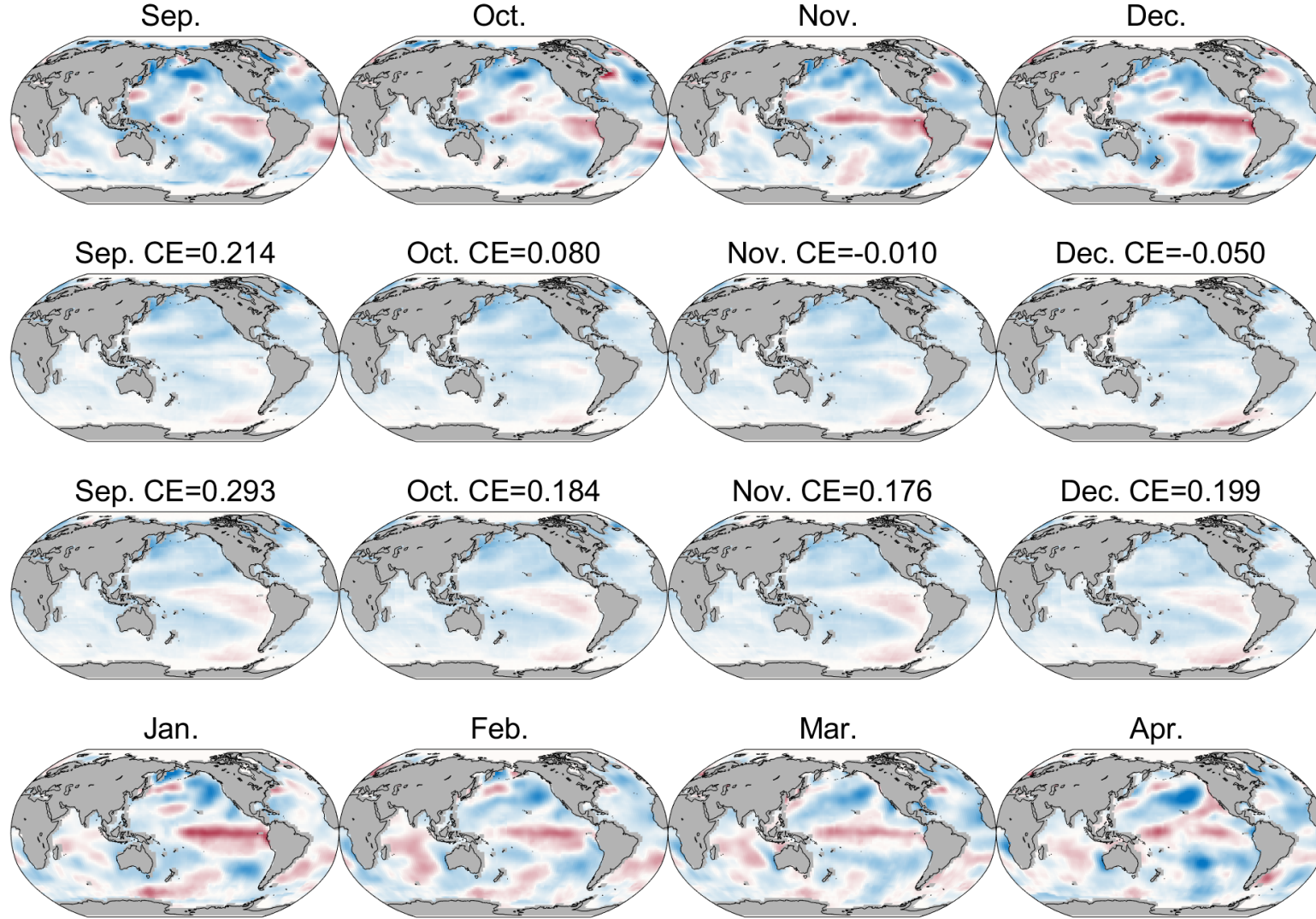
Reconstruction snapshot

1913 ERSST

1913 IHEncF

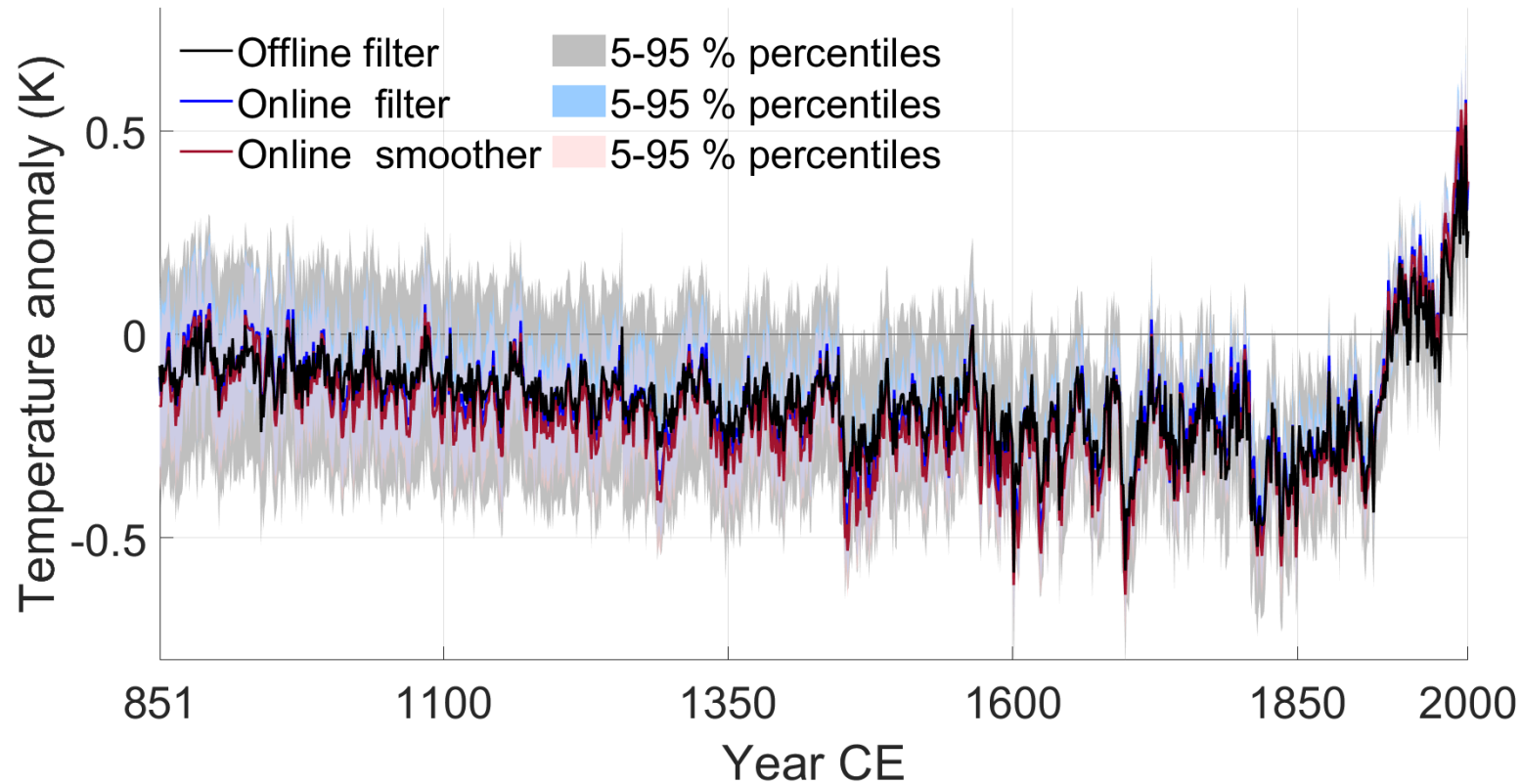
1913 IHEncS_GGF

1914 ERSST



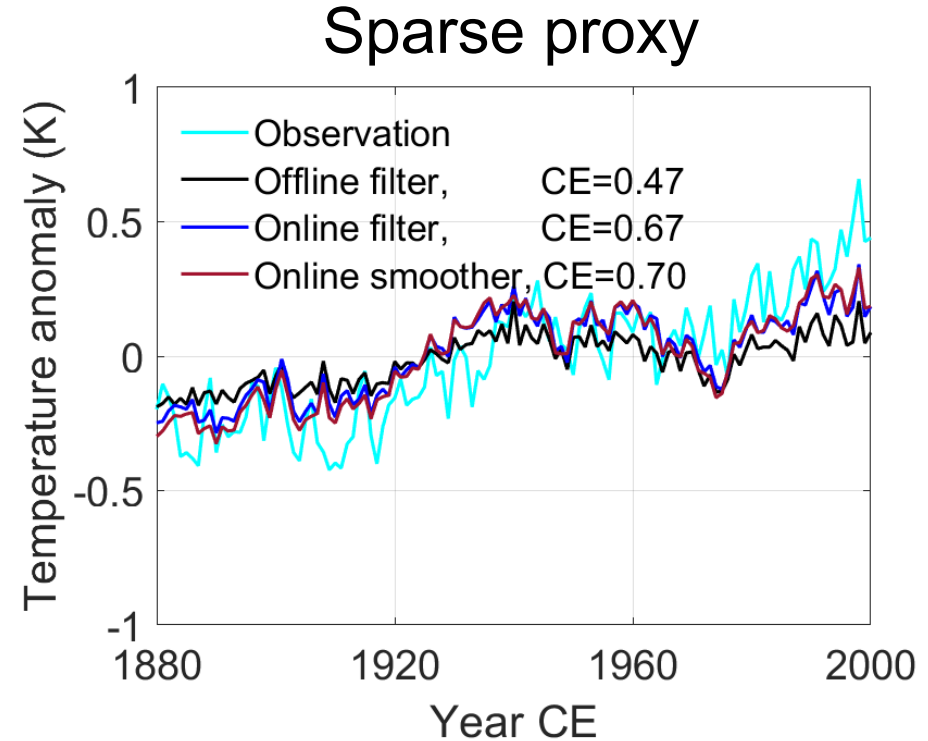
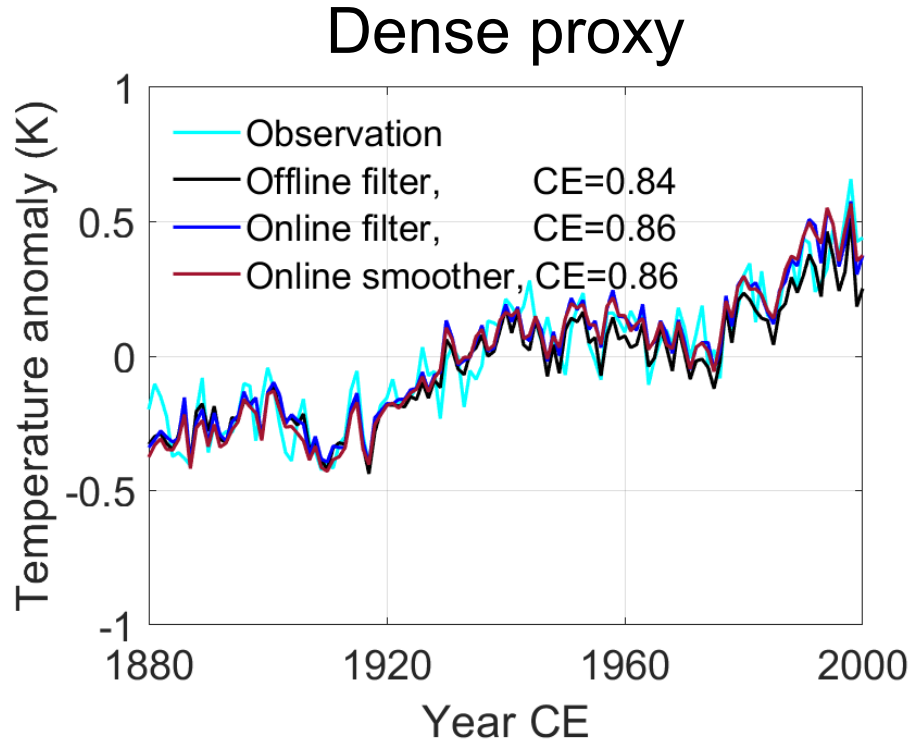
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‘seeing’ the future

Reconstruction of the last 1000 years



- Similar long-term climate evolution across methods
- Larger differences and uncertainties in the early period

Verification in the last 100 years



- PDA gives promising reconstructions with dense proxies
- Online filter shows stronger warming trend with sparse proxies
- Online smoother gives the best performance

Summary

- An EnKS for online paleoclimate data assimilation is proposed, which utilize proxy data from past, current and future
- PDA is traditionally offline, we first **utilize AI model to enable PDA online**. To deal with errors in AI ensemble:
 - ① Blend with climatological ensemble
 - ② Hybrid climatological error covariances
- We second **extend online PDA to smoothing**, by further assimilating future proxy. To improve estimation of cross-time error covariances:
 - ① Temporal localization
 - ② Hybrid climatological cross-time error covariances
- EnKS is superior to that online EnKF and traditional offline EnKF to reconstruct the temperature

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Discussions

- 1. Proxy → Observation operator & observation error**
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2. AI model → Model error & uncertainty propagation
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- ...



Paleoclimate involves multiple timescales of the coupled Earth system

PDA is a testbed for DA under sparse observations, imperfect models, and multiple timescales.

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